

Honeywell

GUS
Faceplate, Alarm, and Message
Scripting
User's Guide

GU2A-400

R400

10/2011

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About This Document

This document contains information for using the GUS Alarm window and summary display. It also includes information on configuring a faceplate control.

Release Information

Document Name	Document ID	Release Number	Publication Date
Faceplate, Alarm, and Message Scripting User's Guide - GU2A	GU2A400	400	10/2011

Document Category
Configuration

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Document ID
<i>GUS Display Builder User's Guide</i>	GU23
<i>Display Scripting User's Guide</i>	GU24
<i>TPS System Administration Guide</i>	TP06X
<i>TPS System Operations Guide</i>	TP07X
<i>TPS System Implementation Guide</i>	TP08X
<i>TPS System Planning Guide</i>	TP10X
<i>TPN Server User's Guide</i>	TP11X
<i>OPC Data Access Specification Reference Manual</i>	TP40
<i>HCI Client Developer's Guide</i>	TP60
<i>PHD OPC Server User's Guide</i>	PIM2901

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE - EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation, which if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation, which if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.

Symbol Definitions

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1. GUS Alarm Window Control

1.1 GUS Alarm Window Control Overview

Functional Description

The Alarm Window Control is a Windows control for displaying and operating TPS Network alarms on a GUS node at runtime. The control connects to an alarm server, executing on Windows XP, to get its collection of alarm data. The alarm server retrieves and caches alarm events from the TPS Network through custom load modules. The alarm server also provides alarm status information (such as the number of area alarms, alarm status, and unit status) to the Alarm Window Control through OLE Automation methods.

Properties and Methods

The Alarm Window Control provides both properties and methods, accessible through property pages and scripting, allowing manipulation of various components of the data displayed in the Alarm Window. Most Alarm Window Control properties have both read and write capabilities and are available at buildtime and runtime. Alarm Window Control properties give you the ability to configure the runtime display. A user may customize the content, font, data view, sorted order, and operation for the alarm data. Additionally, a user may select a set of properties for change notification. If the value of the property changes automatically or through user intervention, you will be notified.

The Alarm Window Control provides several methods to view and operate alarms for paging, acknowledgement, and event recovery. It also provides the ability to call up the Detail point or Associated display of a selected alarm in the Native Window.

GUS Alarm Window ActiveX control not supported on Experion

The GUS Alarm Windows ActiveX control is not supported on ES-T nodes. Existing GUS displays that are embedding this control will present a standard “control not installed” error if the GUS display is invoked from an Experion node. Additionally, operators attempting to request a GUS Alarm Summary from an Experion Station will instead see the Experion Alarm Summary appear. The GUS Alarm Window ActiveX control is not included in the Experion installation packages and the supporting TPN load modules (GUSALA, GUSALAG) do not need to be configured for ES-T nodes.

1 GUS Alarm Window Control

1.2 GUS Alarm Window Control - Buildtime Mode

Buildtime and Runtime Modes

The Alarm Window Control operates differently in its two modes: buildtime and runtime. The buildtime mode provides property pages to allow you to size and configure the type of alarm data to be displayed at runtime.

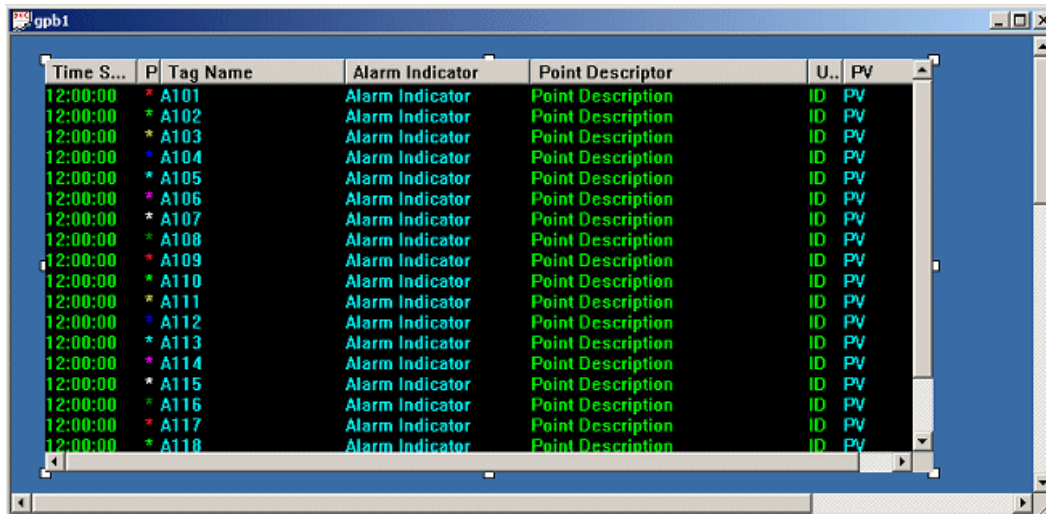
1.2 GUS Alarm Window Control - Buildtime Mode

Buildtime Appearance

The following figure represents the Alarm Window Control at buildtime.

The dummy text displayed in the Alarm Window Control is provided as an aid to customize the position and layout of the alarm data. The dummy data is only a sample – in some cases, it may not provide the entire field width necessary to show all the live data (for example, a TPN system configured for 16 character tags names would need a wider Tag Name field).

Figure 1 Buildtime Alarm Window Control



The screenshot shows a window titled 'gpb1' containing a table with the following columns: Time S..., P, Tag Name, Alarm Indicator, Point Descriptor, U., and PV. The table contains 18 rows of dummy data, all with a time of 12:00:00. The 'Tag Name' column contains tags A101 through A118. The 'Alarm Indicator' column contains the text 'Alarm Indicator'. The 'Point Descriptor' column contains the text 'Point Description'. The 'U.' column contains 'ID' and the 'PV' column contains 'PV'.

Time S...	P	Tag Name	Alarm Indicator	Point Descriptor	U.	PV
12:00:00	*	A101	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A102	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A103	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A104	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A105	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A106	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A107	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A108	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A109	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A110	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A111	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A112	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A113	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A114	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A115	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A116	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A117	Alarm Indicator	Point Description	ID	PV
12:00:00	*	A118	Alarm Indicator	Point Description	ID	PV

Buildtime Property Pages

Multiple property pages are available at buildtime to configure different aspects of the Alarm Window Control, including features such as column visibility, window size, font, and line count.

Buildtime Column Headings

At buildtime, the column headings of the Alarm Window Control occupy the same position as a Window's title bar and indicate the alarm line Field Names.

You can change the columns at buildtime as follows:

- Resize the width of any column by dragging the slit between fields. The column widths you modify at buildtime will be persisted at runtime.
- Make the column headings visible or invisible, through the property page.
- Choose which columns of data are displayed, through the property page.

Buildtime Alarm Content

At buildtime, the Alarm Window Control provides a tabular display of dummy data. The data is organized by field names in each column.

Property pages allow you to affect the alarm window content as follows:

- Enable or disable the visibility of specific columns. (If a column is required, the user cannot alter its associated visibility property.)
- Configure the number of lines per page (1 – 20).
- Configure the font style and color.
- Configure the background color.
- Select either the Time Stamp or Priority sort order for startup.
- Enable or disable the zoom to fit.

Buildtime Scroll Bars

A horizontal or vertical scroll bar is displayed, if needed, to aid you in configuring the size of the Alarm Window.

1.3 GUS Alarm Window Control - Runtime Mode

Runtime appearance

The following figure represents the Alarm Window Control at runtime, collecting data from the TPS Network.

Figure 2 Runtime Alarm Window Controls

Time S...	P	Tag Name	Alarm Indicator	Point Descriptor	U.	PV
11:41:19		AM DAVE2	BADPV	test nar 1-4o5lc	82	50.0
09:42:29	E	B1 IDFAN ...	PVHI 20.0	BOILER 1 ID FAN CONTROL	BB	25.0
09:42:27	H	GB REG	PVHI 5.0	xxxxxxxxxxxxxxxxxxxxxx	BB	95.0
09:42:27	L	AMPVCALC	PVLO 5.0		BB	11.8
09:42:27	E	B1 FDFAN...	PVHI 20.0	BOILER 1 FD FAN CONTROL	BB	11.8
09:42:25	L	545 AM01	BADPV	R600 Journal Expansion	BB	-----
09:42:24	L	AM29MDVP1	BADPV	MULTIPLY/DIVIDE EQ. A	BB	3.3
09:42:24	L	AMTREND1	BADPV	TREND TEST POINT	85	11.8
09:42:24	L	AMTREND1	CLFALM	TREND TEST POINT	85	11.8
09:42:24	L	R6107 DTEC	PVLO 40.0		90	11.8
09:42:24		545 GB	BADPV	cont hist test nt	90	11.8
09:42:24		545 GB	CLFALM	cont hist test nt	90	11.8
09:42:24	L	545 AM09	BADPV	R600 Journal Expansion	BB	-----
09:42:24	L	545 AM07	BADPV	R600 Journal Expansion	BB	-----
09:42:24	L	545 AM05	BADPV	R600 Journal Expansion	BB	-----
09:42:24	L	545 AM02	BADPV	R600 Journal Expansion	BB	-----
09:42:24	L	545 AM03	BADPV	R600 Journal Expansion	BB	-----
09:42:24		PERF9 K4	BADPV	ROLL AV PERF IMPACT K4	8	11.7
09:42:24		PERF8 K4	BADPV	ROLL AV PERF IMPACT K4	8	11.7
09:42:24		PERF6 K4	BADPV	ROLL AV PERF IMPACT K4	8	11.7

Runtime column headings

At runtime, the column headings occupy the same position as a Window's title bar, indicating the Field Names.

At runtime, the user can modify the alarm window content as follows:

- Sort the alarms by Time Stamp or Priority order by clicking the field name in the Column Heading.
- Resize the width of a column by dragging the slit between fields. (The column widths you modify at runtime will NOT be persistent at buildtime.)

Runtime alarm content

At runtime, the Alarm Window Control provides a tabular display organized by field names. The following table describes the Alarm Window Control runtime fields.

Table 1-1 Alarm Window Control – Field Descriptions	
COLUMN	DESCRIPTION
Time Stamp	Indicates what time, date, or time and date the alarm was detected. When sorted by timestamp, the latest alarms appear on the top.
Priority	Indicates the priority of the alarm that occurred (Emergency, High or Low), represented by symbols or characters and alarm color, as defined in the system's NCF configuration: Square or E = Emergency Up triangle or H = High Down triangle or L = Low) The priority indicator flashes until the alarm is acknowledged. The priority indicator is reverse video/half intensity white if the alarm is unacknowledged and the alarm condition has cleared. When sorted by alarm priority, the order of sorting is as follows: Unacknowledged Emergency Acknowledged Emergency Unacknowledged High Acknowledged High Unacknowledged Low Acknowledged Low Within the above categories, alarms are also sorted by reverse chronological order.
Tag Name	Indicates the tag name at which the alarm occurred.
Alarm Indicator	Indicates the reason for the alarm and the exceeded limit, if applicable.

1 GUS Alarm Window Control

1.3 GUS Alarm Window Control - Runtime Mode

Table 1-1 Alarm Window Control – Field Descriptions	
COLUMN	DESCRIPTION
Point Descriptor (Or Primary Module or Multiple Primary Module)	Indicates the alarm's Point Descriptor. You can switch the data displayed in this column by modifying the ChangeDescription property. The data in the column can be switched between the Point Descriptor, Primary Module, and Multiple Primary Module (only if the corresponding column is not configured to be displayed separately).
Primary Module	Indicates the Primary Module of the point at which the alarm occurred.
Multiple Primary Modules 1 - 4	Indicates Multiple Primary Modules of the point at which the alarm occurred.
Unit ID	Indicates the Unit ID of the point at which the alarm occurred.
PV (or Engineering Units Description)	Indicates the alarm's current PV value. The data in this column can be switched between the PV Value and the Engineering Units Description on the GUS Alarm Summary.

Runtime view filter

The user can classify alarms (Area or Unit) shown on the Alarm Window Control by the View Filter property accessible from a Property Page. The following table describes each view filter.

Table 1-2 Alarm Window Control - View Filter		
VIEW FILTER	CATEGORY	VALUE
Area Alarm	All alarms that the station received	0
Unit Alarm	Alarms that belong in the specified Unit	1-36

Runtime scroll bars

A horizontal or vertical scroll bar is displayed on the Alarm Window Control, if needed. The vertical scroll bar allows the user to scroll alarms within a current page (not over several pages).

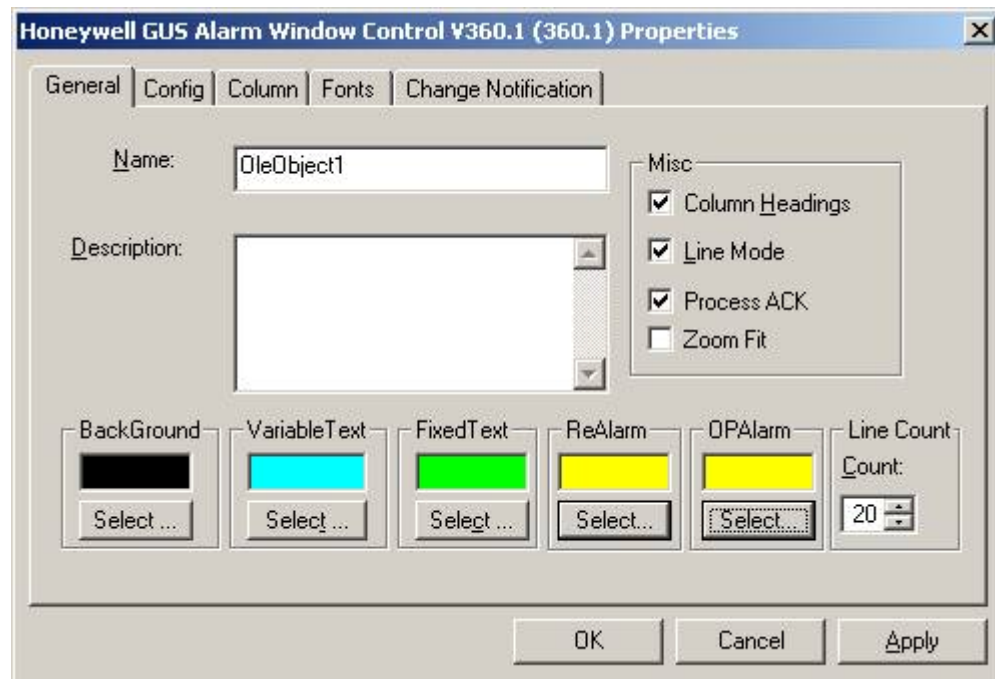
When the vertical scroll bar is displayed, Page-Acknowledge may acknowledge alarms that are not displayed.

1.4 GUS Alarm Window Control Property Pages

General property page

The General property page contains the most general properties for the Alarm Window Control. The following figure shows the property page.

Figure 3 General Property Page



1 GUS Alarm Window Control

1.4 GUS Alarm Window Control Property Pages

General Properties

The following table describes the properties included on the General properties page of the Alarm Window Control.

Table 1-3 General Property Page	
PROPERTY	DEFINITION
Name	The Name field attaches a user-defined name to the control.
Description	This property attaches a documentation note to the control.
 ATTENTION Name and Description are generic attributes common to all Honeywell custom controls.	
Column Headings	The Column Headings Property allows you to decide whether or not to display the Column Headings section. When displayed, the Column Headings supply functionality to do the following: • Sort alarms by clicking on a field name. • Resize a field width by dragging the field separators.
Line Mode	The Line Mode property allows you to operate in Line Mode as well as Page Mode. When the Control has its mode property set to Line Mode and you select a single alarm, you can acknowledge only that alarm. If no alarm is selected, you can acknowledge the entire page. In addition, operating in Line Mode allows you to invoke the related Native Window Detail Display or Associated Display. The valid range is 1 to 20.
 ATTENTION The display may have multiple pages. If the Line Mode property is not checked, the control operates in Page Mode. When you acknowledge alarms, the acknowledgement applies to the entire page. This may include alarms that are not currently visible due to the presence of the scroll bar.	

Table 1-3 General Property Page	
PROPERTY	DEFINITION
Process ACK	The Process ACK property allows you to determine if the Integrated keyboard (IKB) or Operator Entry Panel (OEP) ACK key event (or QUE_KEY (ACK)) should be processed by the control. When this property is set to ON, the Alarm Window Control will process the IKB/OEP ACK key when the control has focus, and will not forward the key event to the display for further processing. If this property is set to OFF, and the control has focus, the Integrated keyboard (IKB) or Operator Entry Panel (OEP) ACK key event will NOT be processed by the control. This is provided to allow the IKB/OEP ACK key event to be forwarded to the display for processing. The default value is set to ON to support migration from previous releases.
 ATTENTION The Process ACK property does not affect the ability to call the ACK() method on the Alarm Window Control at any time. It is only used to control the processing of keystrokes or simulated keystrokes by the QUE_KEY actor.	
Line Count	The Line Count property controls the number of alarms that appear on a page of the display. If the size of the display is smaller than the Line Count, a vertical scroll bar scrolls the alarms within a single page.
Background	Determines if the background color should be changed. The default value is Black.
VariableText	Determines if the variable text color should be changed. The color changes for the Tag Name, Alarm Indicator and PV column. The default value is Cyan.
FixedText	Determines if the fixed text color should be changed. The color changes for the Time stamp, Point Descriptor and Unit ID columns. The default value is Green.
ZoomFit	Allows you to re-scale the alarm OCX at runtime. The default setting is disabled.

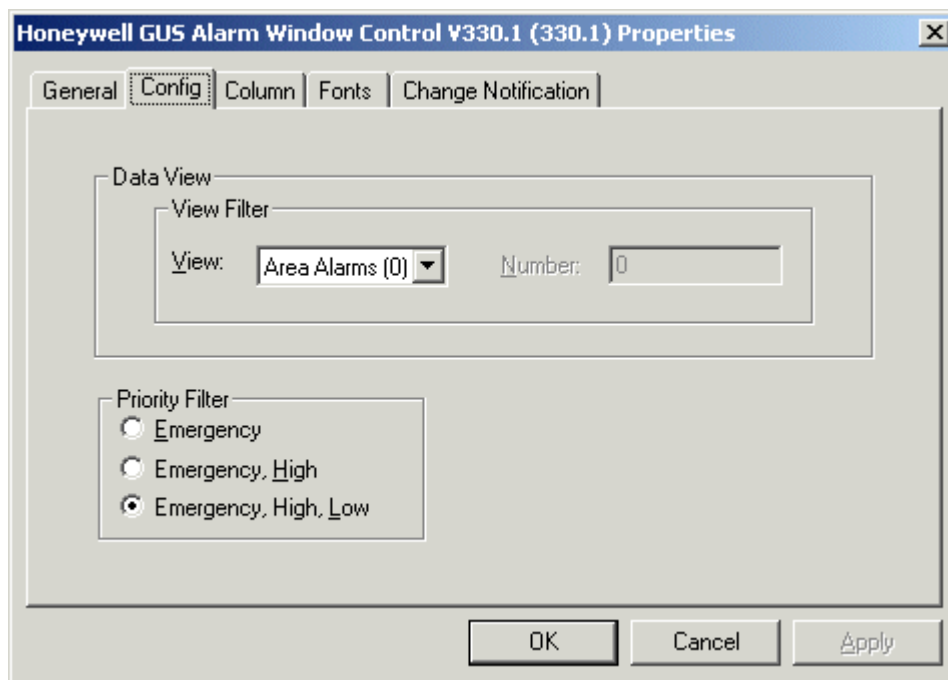
1 GUS Alarm Window Control

1.4 GUS Alarm Window Control Property Pages

Configuration property page

The Configuration (Config) property page contains the properties that configure the filters for the alarm data and the display page. You may filter the alarms by Priority and by Area or Unit. The following figure shows the property page.

Figure 4 Configuration Property Page



Configuration properties

The following table describes the properties included on the Configuration properties page of the Alarm Window Control.

Table 1-4 Configuration Property Page	
PROPERTY	DEFINITION
Data View Group	The Data View Group controls the data content appearing in the Alarm Collection Data section of the Alarm Window Control. The view may be either Area (0) or a specified Unit (1-36).
Priority Filter Group	The Priority Filter Group determines the type of alarms that will be displayed in the Alarm Collection Data section of the Alarm Window Control. You may choose to view the following: • All alarms (emergency, high and low), • Just the emergency and high alarms, or • Only the emergency alarms.

1 GUS Alarm Window Control

1.4 GUS Alarm Window Control Property Pages

Column property page

The Column property page controls the visible columns and the column providing the sorted order for the alarms. The following figure shows the property page.

Figure 5 Column Property Page

The screenshot shows the 'Honeywell GUS Alarm Window Control V330.1 (330.1) Properties' dialog box with the 'Column' tab selected. The dialog has five tabs: General, Config, Column, Fonts, and Change Notification. The 'Column' tab contains the following settings:

- Display Required:** A text box containing '80' and a checked checkbox for 'Time Stamp'.
- Optional:** A group box containing several settings:
 - Tag Name:** A text box containing '160' and a checked checkbox.
 - Alarm Indicator:** A text box containing '160' and a checked checkbox.
 - Priority:** A text box containing '20' and a checked checkbox.
 - PV/EU Desc:** A text box containing '90' and a checked checkbox.
 - Unit ID:** A text box containing '30' and a checked checkbox.
 - Point Descriptor:** A text box containing '240' and a checked checkbox.
 - Primary Module:** A text box containing '160' and an unchecked checkbox.
 - Multiple Primary Module:** A text box containing '160' and an unchecked checkbox.
- Sort Order:** A section with a 'Field:' label and a dropdown menu currently showing 'Time Stamp'.
- Default Column Widths:** A button.

At the bottom of the dialog are three buttons: 'OK', 'Cancel', and 'Apply'.

Column properties

The following table describes the properties included on the Column properties page of the Alarm Window Control.

Table 1-5 Column Property Page	
PROPERTY	DEFINITION
Display Group	The Display Group allows you to decide which fields are displayed in the Alarm Collection Data section of the Alarm Window Control. Grayed out fields indicate that you cannot change the option for that property. A check in the check box associated with the field name indicates that the column will be displayed.
ATTENTION: The following properties support the Column Property Page. For complete details on the properties, see the section 1.5 GUS Alarm Window Control Properties.	
TimeStampColumn	Property that determines whether the Time Stamp column is displayed or not. ATTENTION: You cannot set this property. The default value is set to enabled.
TagNameColumn	Property that determines whether the Tag Name column is displayed or not. ATTENTION: You cannot set this property. The default value is set to enabled.
AlarmIndicatorColumn	Property that determines whether the Alarm Indicator column is displayed or not. ATTENTION: You cannot set this property. The default value is set to enabled.
PriorityColumn	Property that determines whether the Alarm Priority column is displayed or not.
PV/EU DescColumn	Property that determines whether the PV value or Engineering Unit Description (EU Desc) is displayed in the PV column or not.

1 GUS Alarm Window Control

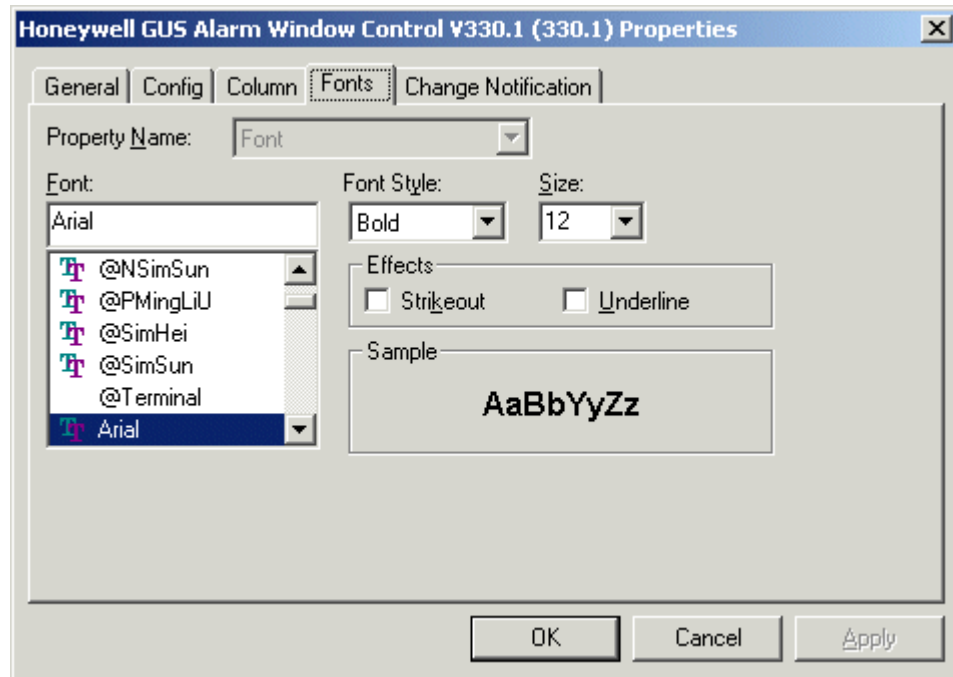
1.4 GUS Alarm Window Control Property Pages

Table 1-5 Column Property Page	
PROPERTY	DEFINITION
UnitIDColumn	Property that determines whether the Unit ID column is displayed or not.
PointDescriptorColumn	Property that determines whether the Point Descriptor column is displayed or not.
PrimmodColumn	Property that determines whether the Primary Module column is displayed or not.
MultiplePrimmodColumn	Property that determines whether the Multiple Primary Module column is displayed or not.
SortItem	Property that identifies the field used to sort the alarms. Values include Time Stamp and Priority.

Fonts Property Page

The Fonts property page controls the font type, style, and size. This is the standard Font page available to any Windows application and is not discussed in detail here.

Figure 6 Fonts Property Page



1 GUS Alarm Window Control

1.4 GUS Alarm Window Control Property Pages

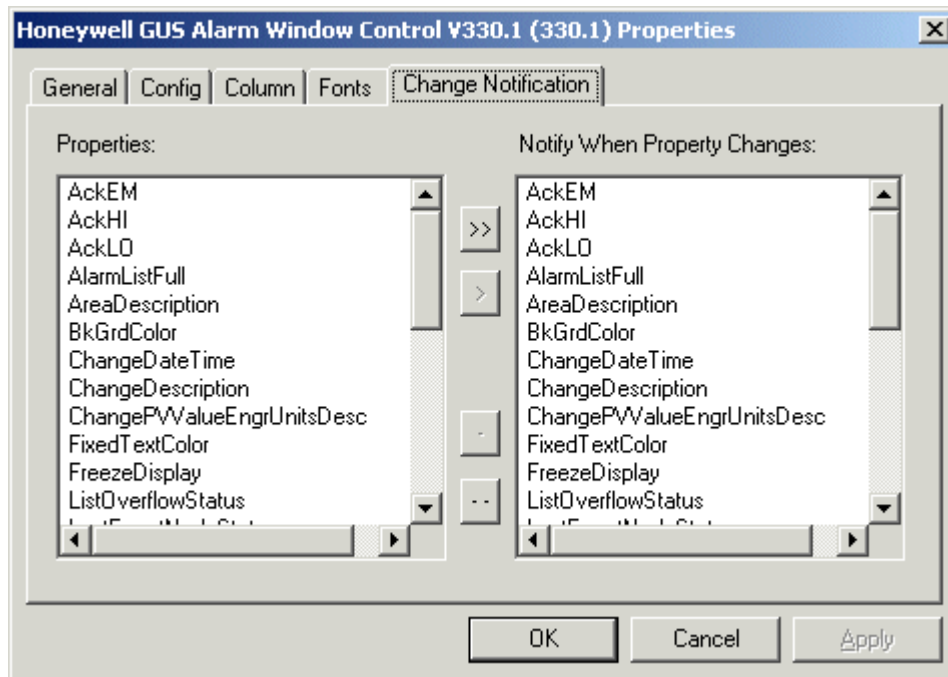
Change Notification property page

The Change Notification property page allows you to select properties that will activate property change events. The following figure shows the property page.

This property page is closely coupled with the scripting function, `OnPropertyChange()`.

The `OnPropertyChange()` event is executed each time the value of one of the selected properties on the property page changes. You cannot change the notification of events at runtime.

Figure 7 Change Notification Property Page



Properties listbox

The *Properties* listbox, on the left side of the Change Notification property page, lists all of the Alarm Window Control runtime properties that provide notification when the property value changes.

Notify when property changes listbox

The *Notify When Property Changes* listbox, on the right side of the Change Notification property page, displays the properties you chose to provide notification when the property value changes.

Selecting properties for property change events

Four buttons appear between the listboxes on the Change Notification property page. Use the buttons as described in the following table.

Use this button...	To do this...
>>	Place all of the properties in the <i>Notify When Property Changes</i> listbox.
--	Remove all properties from <i>Notify When Property Changes</i> listbox.
>	Add the selected property in the <i>Properties</i> listbox to the <i>Notify When Property Changes</i> listbox.
-	Delete the selected property from the <i>Notify When Property Changes</i> listbox.

For Multiple Selections:

Use the Shift key and the mouse for multiple selections of contiguous properties in either listbox.

Use the Ctrl key and the mouse for multiple selection of non-contiguous properties.



ATTENTION

Notification of all properties that are unit specific (for example, `UnitStatus(UnitIndex)`) applies to all units. You cannot specify property change notification for a subset of units.

1 GUS Alarm Window Control

1.5 GUS Alarm Window Control Properties

Change notification runtime operation



ATTENTION

The first OnPropertyChange() event will not be fired in the GUS Display Builder until all of the OnDisplayStartup() scripts have completed. Note that each property change event will fire once when the properties change from uninitialized to initialized.



ATTENTION

A TPN Area Change event causes the properties to go from their previous values to uninitialized, and then to their new values for the new Area.

1.5 GUS Alarm Window Control Properties

Overview

The following tables describe the characteristics of each Alarm Window Control property:

- Table 1-6 Alarm Window Control Properties-Column/Column Headings Visibility
- Table 1-7 Alarm Window Control Properties-Operational/Informational Properties
- Table 1-8 Alarm Window Control Properties-Unit Status/Informational Properties

Property change notification



ATTENTION

For most of the Alarm Window Control properties, user changes performed at runtime (through scripting) result in immediate modification of the property. Some properties require verification of the requested change by the TPS Network, before the Alarm Window Control can set the property value.

The properties requiring verification are not immediately updated with the requested value.

You should use the OnPropertyChange() event to receive notification when the property has been changed. This is especially important for a property change that is enforced by the TPS Network, such as AckEM and UnAckEM.

Column and Column Heading Visibility Properties

The following table defines the Alarm Window Control properties that define the visibility of the column and column headings:

Table 1-6 Alarm Window Control Properties- Column/Column Headings Visibility					
Property	Description	Default Value	Script Type	Configurable Through Property Page	Scripting Access
AlarmIndicatorColumn	Runtime read-only property that determines whether the Alarm Indicator column is displayed or not.	True	Boolean	Yes	Read only
AlarmPriorityColumn	Runtime read-only property that determines whether the Alarm Priority column is displayed or not.	True	Boolean	Yes	Read only
ColumnHeading	Runtime read-only property that determines whether the column headings are displayed or not.	True	Boolean	Yes	Read only
MultiplePrimmodColumn	Runtime read-only property that determines whether the four Multiple Primary Module columns are displayed or not.	False	Boolean	Yes	Read only
PointDescriptionColumn	Runtime read-only property that determines whether the Point Descriptor column is displayed or not.	True	Boolean	Yes	Read only

1 GUS Alarm Window Control

1.5 GUS Alarm Window Control Properties

Table 1-6 Alarm Window Control Properties- Column/Column Headings Visibility					
Property	Description	Default Value	Script Type	Configurable Through Property Page	Scripting Access
PrimmodColumn	Runtime read-only property that determines whether the Primary Module column is displayed or not.	False	Boolean	Yes	Read only
TagNameColumn	Runtime read-only property that determines whether the Tag Name column is displayed or not.	True	Boolean	Yes	Read only
TimeStampColumn	Runtime read-only property that determines whether the Time Stamp column is displayed or not.	True	Boolean	No	Read only
UnitIDColumn	Runtime read-only property that determines whether the Unit ID column is displayed or not.	True	Boolean	Yes	Read only
PV/EU Desc Column	Runtime read-only property that determines whether the PV value or Engineering Unit Description (EU Desc) is displayed in the PV column or not.	True	Boolean	Yes	Read only

Operational and informational properties

The following table defines the operational and informational properties of the Alarm Window Control.

Table 1-7 Alarm Window Control Properties- Operational/Informational Properties					
Property	Description	Default Value	Script Type	Can be configured Through Property Page	Script Access
BkgGrdColor	Property that determines the back ground color	0/TDC_Black (0/RGB(0,0,0))	Long integer	Yes	Read/Write
ChangeDateTime	Property that determines whether the time, date, or both is displayed in the time stamp column. Time= "TIME" Date= "DATE" Time and Date= "BOTH"	"TIME"	String	No	Read/Write
ChangeDescription	Property that determines whether the Point Descriptor, Primary Module Point, or Multiple Primary Module Point information is displayed in the original Point Descriptor column. Point Descriptor = "DESC" Primary Module = "PRIM" Multiple Primary Module = "MPRM" <i>NOTE: This property may only be changed to PRIM or MPRM if the Primary Module Column and/or the Multiple Primary Module Columns are not configured.</i>	"DESC"	String	No	Read/Write

1 GUS Alarm Window Control

1.5 GUS Alarm Window Control Properties

Table 1-7 Alarm Window Control Properties- Operational/Informational Properties					
Property	Description	Default Value	Script Type	Can be configured Through Property Page	Script Access
ChangePVValueEngrUnitsDesc	Property that determines whether the PV Value or the Engineering Units Description is displayed in the PV column. PV Value= "PV" Engr Units Desc = "EUD"	"PV"	String	No	Read/Write
FixedTextColor	Property that determines the fixed text color	65280/TDC_Green (FF00/RGB(0, 255,64))	Long integer	Yes	Read/Write
LineCount	Runtime read-only property that determines how many alarms is displayed per page (1-20).	20	Integer	Yes	Read only
LineOperation	Runtime read-only property that determines whether an operation is done only for the selected Alarm. False = Page operation only True = Single or Page operation	False = Page	Boolean	Yes	Read only
Name	Runtime read-only property that identifies the object name for the control.	N/A	String	Yes	Read only
Page	Runtime read-only property that identifies the page currently being displayed in the alarm summary. The PageForward() and PageBack() methods are used to change the page.	1	Integer	No	Read only

Table 1-7 Alarm Window Control Properties- Operational/Informational Properties					
Property	Description	Default Value	Script Type	Can be configured Through Property Page	Script Access
ProcessAck	The Process ACK property allows you to determine if the IKB/OEP ACK key event (or QUE_KEY (ACK)) should be processed by the control. When this property is set to TRUE, the Alarm Window Control will process the IKB/OEP ACK key when the control has focus, and will not forward the key event to the display for further processing. If this property is set to FALSE, and the control has focus, the IKB/OEP ACK key event will NOT be processed by the control.	True	Boolean	Yes	Read/Write
PriorityFilter	Property that indicates the priority filter on the Alarm Window. "E" = Emergency "EH" = Emergency and High "EHL" = Emergency, High, and Low	"EHL"	String	Yes	Read/Write
SelectedTag	Property that identifies the Tag Name of the selected alarm.	N/A	String	No	Read/Write
SortItem	Property that specifies the column index of the field used for sorting. 1 = Time Stamp 2 = Priority	1 = Time Stamp	Integer	Yes	Read / Write

1 GUS Alarm Window Control

1.5 GUS Alarm Window Control Properties

Table 1-7 Alarm Window Control Properties- Operational/Informational Properties					
Property	Description	Default Value	Script Type	Can be configured Through Property Page	Script Access
VariableTextColor	Property that determines the variable text color	16776960/TDC_Cyan (FFFF00/RGB(0,255,255))	Long integer	Yes	Read/Write
ViewFilter	Property that determines the type of data displayed. 0 = Area 1-36 = Unit Number	0 = Area	Integer	Yes	Read/Write

Status and informational properties

The following table defines the area and unit status and the informational properties of the Alarm Window Control:

Table 1-8 Alarm Window Control Properties- Unit Status/Informational Properties					
Property	Description	Default Value	Script Type	Configurable Through Property Page?	Scripting Access
AreaDescription	Runtime read-only property that identifies the name of the area database in use.	N/A	String	No	Read only
ListOverflowStatus	Runtime read-only property that indicates the status of the list overflow for the node and units. This property is indexed by unit (0 = node, 1-36 = unit). 0 = No overflow 1 = Overflow	0	Integer	No	Read only
LostEventNodeStatus	Runtime read-only property that indicates the status of the lost events and recovery for the node. 0 = No lost events 1 = Lost process events 2 = Lost process network events (Hiway events) 3 = Lost events (both process events and Hiway events) 4 = Recovery in progress	0	Integer	No	Read only

1 GUS Alarm Window Control

1.5 GUS Alarm Window Control Properties

Table 1-8 Alarm Window Control Properties- Unit Status/Informational Properties					
Property	Description	Default Value	Script Type	Configurable Through Property Page?	Scripting Access
LostEventOnThisNode	Runtime read-only property that indicates whether the lost events originated from this node. FALSE = No lost events TRUE = Lost event originated on this node	False	Boolean	No	Read only
LostEventUnitStatus	Runtime read-only property that indicates the status of the lost events and recovery for the unit. This property is indexed by unit number (1-36). 0 = None 1 = Lost events 2 = Lost events notification has been acknowledged 3 = Event recovery requested 4 = Event recovery in progress	0	Integer	No	Read only
UnitDescription	Runtime read-only property that represents the Unit's description. This property is indexed by unit number (1-36).	N/A	String	No	Read only
UnitID	Runtime read-only property that specifies the Unit's ID. This property is indexed by unit number (1-36).	N/A	String	No	Read only

Table 1-8 Alarm Window Control Properties- Unit Status/Informational Properties					
Property	Description	Default Value	Script Type	Configurable Through Property Page?	Scripting Access
UnitStatus	Runtime read-only property that specifies the status of the Unit's alarms. This property is indexed by unit number (1-36). Legal values are: 0 = Not Configured 1 = NOALARM 2 = ACKLO 3 = ACKHI 4 = ACKEM 5 = UNACKLO 6 = UNACKHI 7 = UNACKEM	N/A	Integer	No	Read only
AckEM	Runtime read-only property that specifies the count of acknowledged Emergency alarms. This property is indexed by unit (0 = area, 1-36 = unit).	N/A	Integer	No	Read only
AckHI	Runtime read-only property that specifies the count of acknowledged High alarms. This property is indexed by unit (0 = area, 1-36 = unit).	N/A	Integer	No	Read only

1 GUS Alarm Window Control

1.5 GUS Alarm Window Control Properties

Table 1-8 Alarm Window Control Properties- Unit Status/Informational Properties					
Property	Description	Default Value	Script Type	Configurable Through Property Page?	Scripting Access
AckLO	Runtime read-only property that specifies the count of acknowledged Low alarms. This property is indexed by unit (0 = area, 1-36 = unit).	N/A	Integer	No	Read only
UnAckEM	Runtime read-only property that specifies the count of unacknowledged Emergency alarms. This property is indexed by unit (0 = area, 1-36 = unit).	N/A	Integer	No	Read only
UnAckHI	Runtime read-only property that specifies the count of unacknowledged High alarms. This property is indexed by unit (0 = area, 1-36 = unit).	N/A	Integer	No	Read only
UnAckLO	Runtime read-only property that specifies the count of unacknowledged Low alarms. This property is indexed by unit (0 = area, 1-36 = unit).	N/A	Integer	No	Read only

1.6 GUS Alarm Window Control Methods

The following table describes the methods that support Alarm Window Control runtime operations through scripting.

Table 1-9 Alarm Window Control Methods				
Method	Description	IKB/OEP Key	Parameter Script Type	Return Script Type
Ack	Method used to acknowledge one alarm or one page of alarms. The return value only indicates that the TPS Network <i>accepted</i> the ACK request, it does not indicate the ACK request has been <i>processed</i> by the TPS Network.	ACK NOTE: The GUS Display Builder contains an OnACK() event and an ACK scripting extension to customize processing of the IKB/OEP ACK key.	None	Boolean
AssocDisplay	Method used to request the associated display to the Native Window. If the argument is a non-empty string, it is used as the tag name for the associated display. If an empty string is passed, the current selected item on the display is used.	ASSOC DISP	String	None
DetailPoint	Method used to send a point detail request to the Native Window. If the argument is a non-empty string, it is used as the tag name for the detail point. If an empty string is passed, the current selected item on the display is used.	DETAIL	String	None

1 GUS Alarm Window Control

1.6 GUS Alarm Window Control Methods

Table 1-9 Alarm Window Control Methods				
Method	Description	IKB/OEP Key	Parameter Script Type	Return Script Type
RecoverEvents	Method used to initiate an Event Recovery Procedure. A return value of TRUE only indicates that the TPS Network <i>accepted</i> the request; it does not indicate the TPS Network has processed the request. FALSE will be returned if RecoverEvents() is called while a recovery is in progress.	N/A	Short (0 = area)	Boolean
PageBack	Method used to return to the previous page of alarms.	PAGE BACK	None	None
PageForward	Method used to advance to the next page of alarms.	PAGE FWD	None	None



ATTENTION

The PageBack and PageForward methods are not circular; that is, paging forward from the last page does not wrap to the first page. You can provide circularity through scripting.

1.7 GUS Alarm Window Control OnPropertyChange() Event

Functional Description

OnPropertyChange() is an event that is automatically executed on the Alarm Window Control at runtime when any property changes value. The change can happen automatically, such as when new alarm data is received. The change can also be user-instigated, such as changing the View Filter property. The first OnPropertyChange() event will not occur in the GUS Display Builder until all the OnDisplayStartup() scripts have completed.

Change Notification Property Page

The buildtime property page allows you to select the properties that are to activate events. You cannot change the notification events at runtime.



REFERENCE - INTERNAL

See the Change Notification property page section for details on the Change Notification Property Page.

Scripting syntax

The BasicScript syntax for this event is as follows:

```
OnPropertyChange(nProperty As Integer, nUnitNumber As Integer,  
vValue As Variant)
```

where:

- nProperty is an integer value that is an index indicating the specific property whose value has changed (see Table 1-10 Programmatic Interfaces)
- nUnitNumber is an integer that indicates the unit number, if applicable. If the property is not associated with a unit, then this value is 0
- vValue is a variant that contains the new value of the property that changed at the time the event was sent. (NOTE: You can alternatively reference the Alarm Window Control object explicitly to get the current value of the property.)

1 GUS Alarm Window Control

1.7 GUS Alarm Window Control OnPropertyChange() Event

Scripting constants



ATTENTION

Built-in scripting constants are provided that correspond to the integer value of the property. You must reference only the built-in constants. The property index numbers are not exposed and thereby cannot be referenced.

Table 1-10 Programmatic Interfaces	
Property	Built-In Constant
AckEm	ALARM_ACKEM
AckHi	ALARM_ACKHI
AckLo	ALARM_ACKLO
AreaDescription	ALARM_AREADESCRIPTION
BkgGrdColor	ALARM_BKGRDCOLOR
ChangeDateTime	ALARM_CHANGEDATETIME
ChangeDescription	ALARM_CHANGEDESCRIPTION
ChangePVValueEngrUnitsDesc	ALARM_CHANGEPVVALUEENGRUNITSDISC
FixedTextColor	ALARM_FIXEDTEXTCOLOR
FreezeDisplay	ALARM_FREEZEDISPLAY
ListOverflowStatus	ALARM_LISTOVERFLOWSTATUS
LostEventNodeStatus	ALARM_LOSTEVENTNODESTATUS
LostEventOnThisNode	ALARM_LOSTEVENTONTHISNODE
LostEventUnitStatus	ALARM_LOSTEVENTUNITSTATUS
Page	ALARM_PAGE
ProcessAck	ALARM_PROCESSACK
PriorityFilter	ALARM_PRIORITYFILTER
SelectedTag	ALARM_SELECTEDTAG

Table 1-10 Programmatic Interfaces	
Property	Built-In Constant
SortItem	ALARM_SORTITEM
UnAckEm	ALARM_UNACKEM
UnAckHi	ALARM_UNACKHI
UnAckLo	ALARM_UNACKLO
UnitDescription	ALARM_UNITDESCRIPTION
UnitID	ALARM_UNITID
UnitStatus	ALARM_UNITSTATUS
VariableTextColor	ALARM_VARIABLETEXTCOLOR
ViewFilter	ALARM_VIEWFILTER

1 GUS Alarm Window Control

1.7 GUS Alarm Window Control OnPropertyChange() Event

Scripting Example

The following is an example of the scripted OnPropertyChange() event on the Alarm Window Control, using the index values for the Property. Assume the display contains the following objects:

- An Alarm Window Control object, Alarm1
- Two text objects, Text1 and Text2, that will display the current Page number and UnitStatus properties

It is not necessary for these two text objects to have an OnDataChange() script or Variable Text Object (VTO) to display the value. The value is instead assigned in the OnPropertyChange script.

The Alarm1 object has the following script:

```
Sub OnPropertyChange(nProperty As Integer, nUnitNumber As Integer, vValue
As Variant)
    Select Case nProperty
        Case ALARM_PAGE
            ' Get the value from the event (vValue)
            Text1.text = vValue

        Case ALARM_UNITSTATUS
            If nUnitNumber = 1 then
                ' Get the value from the event (vValue)
                Text2.text = vValue
            elseif nUnitNumber = 2 then
                ' Can also get the value from the property:
                Text2.text = Alarm1.UnitStatus(nUnitNumber)
            End if
        End Select
    End Sub
```

1.8 GUS Alarm Window Control Error Handling

Error indications

When the Alarm Window Control loses communications with the TPS Network, the following errors indicate that the Alarm Window is not collecting data:

- The control replaces its data with a large red X in its window.
- All methods that have return values return an error.
- The values of some of the runtime properties are replaced with invalid data.

When communication with the TPS Network is restored, the Alarm Window Control re-establishes connection to the TPS Network, resumes data collection, and populates the properties with the new data.

Property values upon TPN connection loss

The following table identifies the affected properties and their values when the control does not have communication with the TPS Network.

Table 1-11 Property Value Upon Connection Loss	
Property	Lost Connection Value
AreaDescription	
ListOverflowStatus	-1
LostEventNodeStatus	-1
LostEventOnThisNode	-1
LostEventUnitStatus	-1
UnitDescription	
UnitID	
UnitStatus	-1
AckEM	-1
AckHI	-1

1 GUS Alarm Window Control

1.8 GUS Alarm Window Control Error Handling

Table 1-11 Property Value Upon Connection Loss	
Property	Lost Connection Value
AckLO	-1
UnAckEM	-1
UnAckHI	-1
UnAckLO	-1

1.9 QWERTY Keyboard Operation with GUS Alarm Window Control

The Page Back/Up and Page Fwd/Dn keys on the QWERTY keyboard operate differently on the Alarm Window Control from the Page Fwd and Page Back keys on the Integrated keyboard (IKB) or Operator Entry Panel (OEP) keyboards. This difference is due to standard Microsoft operation for lists with scroll bar ability.

The following table compares the operation of keys on the QWERTY keyboard with the IKB and OEP keyboards. The behavior described in the following table reflects standard Microsoft OCX behavior.

Table 1-12 QWERTY Keyboard Mapping			
Page Back/Page Up QWERTY Keyboard	Page Fwd/Page Dn QWERTY Keyboard	Page Fwd IKB/OEP	Page Back IKB/OEP
If the vertical scroll bar is visible, this key traverses up within the page of alarms that is displayed.	If the vertical scroll bar is visible, this key traverses down within the page of alarms that is displayed.	Pages forward to the next page of alarms.	Pages backward to the previous page of alarms.
(Line Mode) Automatically selects the top-most alarm displayed, regardless of if the scroll bar is visible. (Page Mode) No alarm is selected.	(Line Mode) Automatically selects the bottom-most alarm displayed, regardless of if the scroll bar is visible. (Page Mode) No alarm is selected.	No alarm is selected.	No alarm is selected.



ATTENTION

If an IKB or OEP is unavailable, the Alarm Window Control provides PageForward and PageBack methods (available through scripting) to achieve the same functionality as the Page Fwd and Page Back keys on the IKB and OEP.

1 GUS Alarm Window Control

1.9 QWERTY Keyboard Operation with GUS Alarm Window Control

2. GUS Alarm Summary Display

2.1 GUS Alarm Summary Display Overview

Functional Description

In general, the GUS Area and Unit Alarm Summaries show alarm data in the same manner as the standard Native Window Area and Unit Alarm Summaries. In addition, operator interaction with both the Native Window and GUS Alarm Summaries is very similar.

It is assumed that the reader is familiar with standard Native Window Alarm Summary functionality; therefore, only the specific differences between the standard Native Window and GUS Alarm Summaries are noted in this section.



REFERENCE - INTERNAL

For a more detailed description of the Native Window Alarm Summary functionality, refer to the Native Window alarm documentation:

TPN Bookset #3050 Operations/Process Operations
Process Operations Manual
SW11-501 (for release 530)
SW11-601 (for release 630 and later)



ATTENTION

The functional descriptions that follow apply to both the GUS Area Alarm Summary as well as the GUS Unit Alarm Summary displays, unless otherwise noted.

GUS Alarm Window ActiveX control not supported on Experion

The GUS Alarm Windows ActiveX control is not supported on ES-T nodes. Existing GUS displays that are embedding this control will present a standard “control not installed” error if the GUS display is invoked from an Experion node. Additionally, operators attempting to request a GUS Alarm Summary from an Experion Station will instead see the Experion Alarm Summary appear. The GUS Alarm Windows ActiveX control is not included in the Experion installation packages and the supporting TPN load modules (GUSALA, GUSALAG) do not need to be configured for an Experion Station node that supports TPS options.

Display resolution and size

The GUS Alarm Summary Display has been designed for two resolutions:

- 1280 x 1024 (default), and
- 1024 x 768.

If you want to change the PC desktop area to the alternate setting you should also edit your copy of the Alarm Summary display's size property in order to keep the proportions correct.

A PC desktop setting of 1024 x 768 requires a display size of 1012 x 718.
Likewise, a PC desktop size of 1280 x 1024 requires a display size of 1014 x 779.

You should maintain the same aspect ratio when specifying a new size.

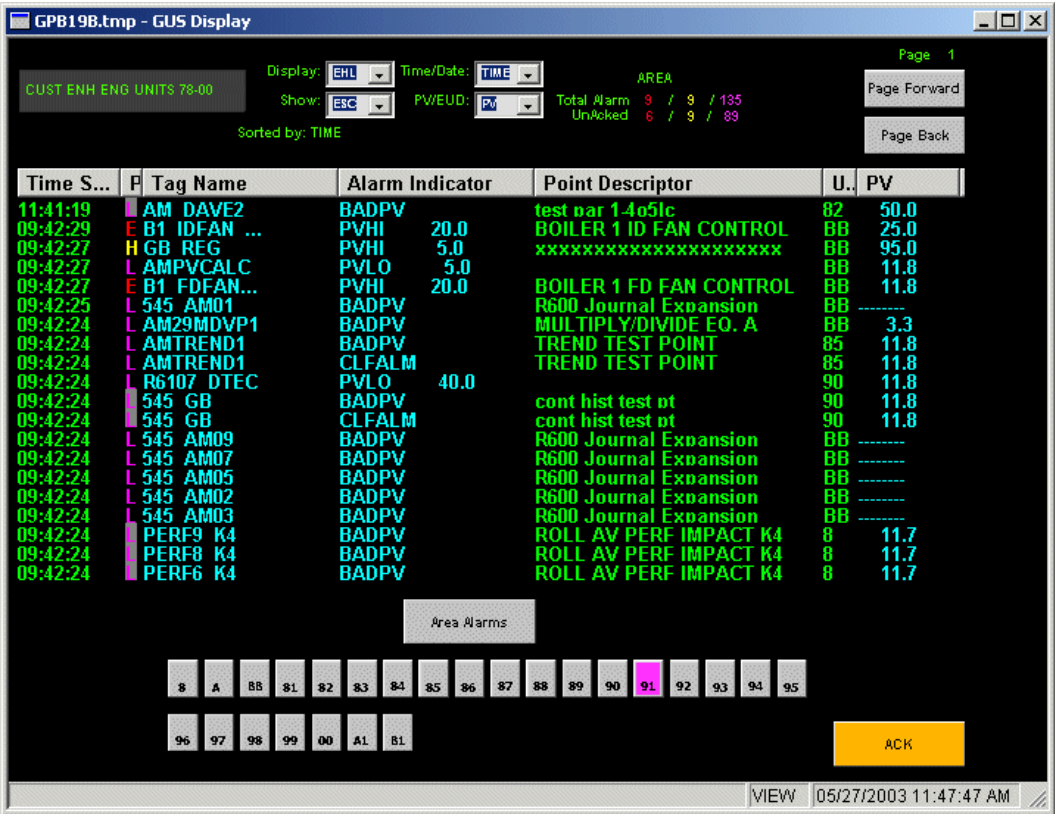
Display examples

The following two figures provide an over-all view of these GUS Alarm Summary displays:

- Area Alarm Summary Display sorted by TIME
- Unit Alarm Summary sorted by TIME

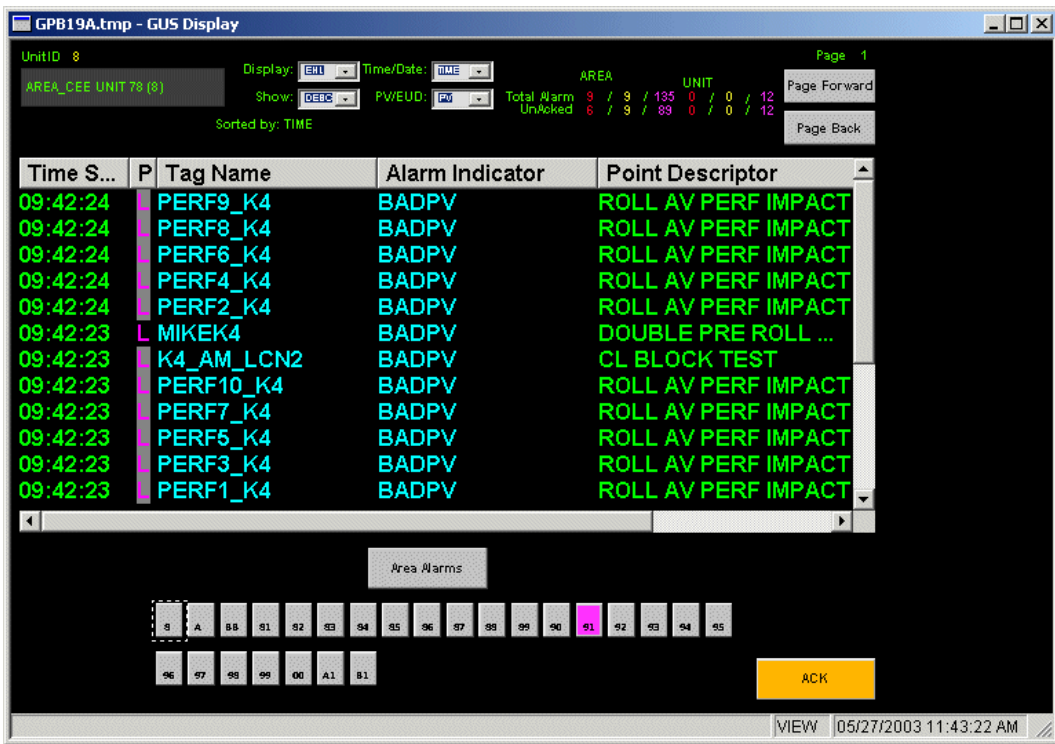
(NOTE: Upon invocation, the GUS Alarm Summary defaults to the AREA view.)

Figure 8 GUS Area Alarm Summary Display Sorted by TIME



2 GUS Alarm Summary Display
2.1 GUS Alarm Summary Display Overview

Figure 9 GUS Unit Alarm Summary Display Sorted by TIME



2.2 GUS Alarm Summary Header Information

Header definitions (top of GUS Alarm Summary)

The following paragraphs describe the header information appearing at the top of the GUS Area and Unit Alarm Summaries.

Area and Unit Descriptors

As shown in the following figure, the Area Alarm Summary display shows the Area Descriptor in the upper left-hand corner and the Unit Alarm Summary display shows the Unit Descriptor and Unit ID.

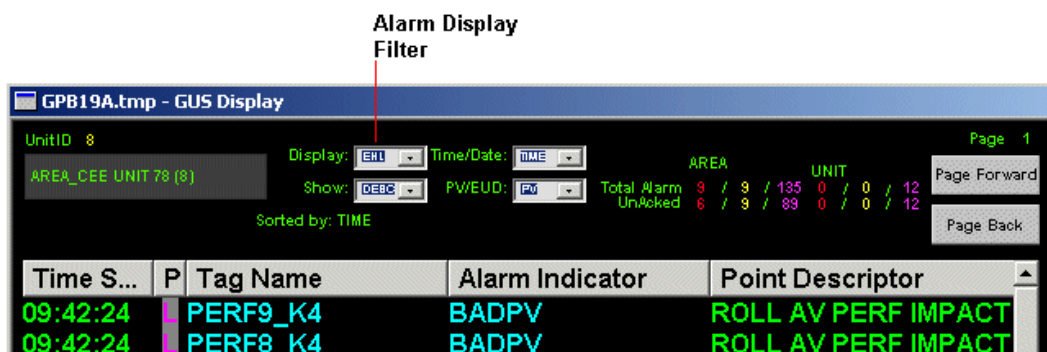
Figure 10 Area and Unit Descriptors



Alarm Display Filter (Display) Pull-Down Menu

The **Display:** pull-down menu corresponds to the “On DISP:” target located on the Native Window Area and Unit Alarm Summary displays. From the pull down menu, you can select **E**, **EH**, or **EHL**. The default is **EHL**.

Figure 11 DISPLAY Pull-Down Menu



ATTENTION

GUS does not read the NCF option that prevents Low Priority alarms from being displayed in the Area Alarm Summary; therefore, if you do not wish to see Low Priority alarms in the Area Summary, you must select the **EH** option from the pull-down menu.

Since the **EH** selection remains in effect until you choose another setting, the operator must select **EHL** from the pull-down menu in order to view Low Priority alarms on a Unit Alarm Summary display.

DESC/PRIM/MPROD (Show) Pull-Down Menu

The **Show** pull-down menu corresponds to the DESC/PRIM/MPROD target located on the Native Window Area and Unit Alarm Summary displays. The menu choices are the same. Refer to the section Point Descriptor/Primmod/Multiple Primmod Field for a detailed description of how this selection affects data on the alarm line.

Figure 12 SHOW Pull-Down Menu

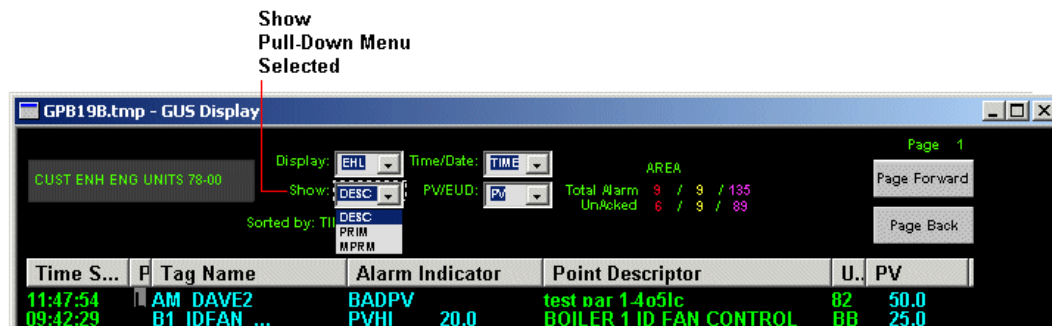
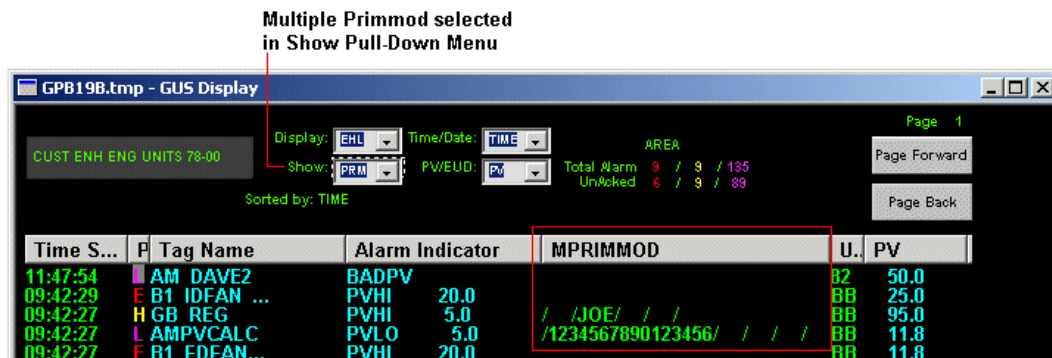


Figure 13 MPRIMMOD Field



2 GUS Alarm Summary Display

2.2 GUS Alarm Summary Header Information

Dynamic Sort Indicator

The text “Sort by:” indicates which dynamic sort method you have selected. The text will read “Sort by: TIME” if the dynamic sorting is being done chronologically. “Sort by: PRIORITY” is displayed if the alarms are being displayed in order of their alarm priority.

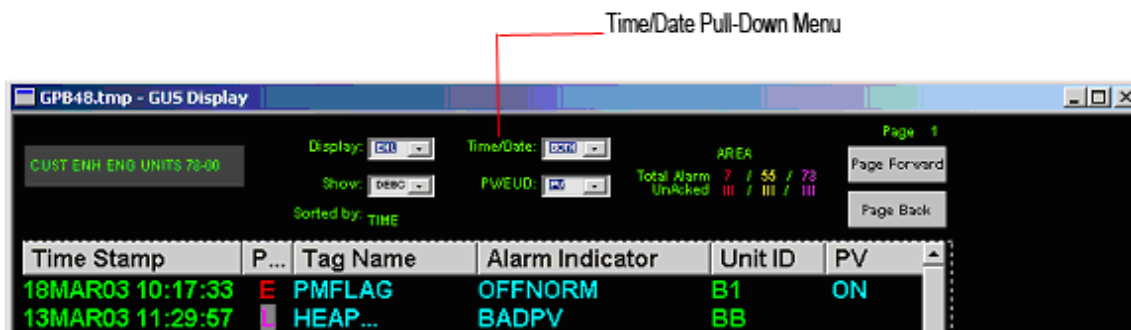
Dynamic Sorting

Alarms may be sorted chronologically or by alarm priority. To choose chronological sorting, select the **Time Stamp** column heading. To sort by alarm priority, select the **Priority** heading.

TIME/DATE/BOTH Pull-Down Menu

The **Time/Date** pull-down menu corresponds to the TIME target on the Native Window Alarm Summary. It allows the user to manually toggle between displaying the time or the date or the time and date in the timestamp field for all alarm lines.

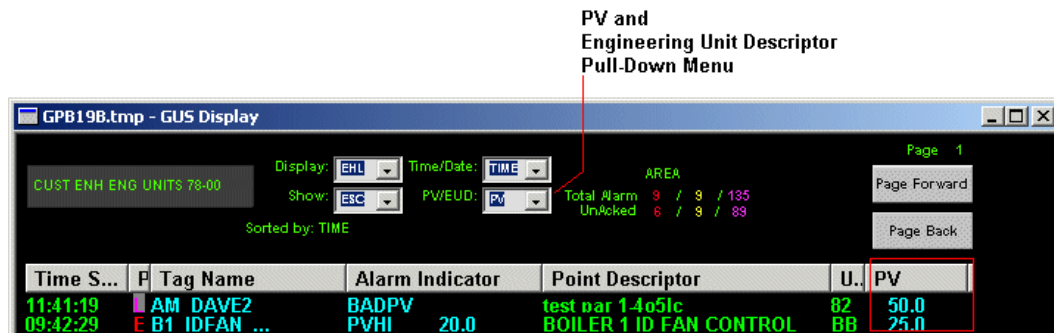
Figure 14 Time/Date Pull-Down Menu



PV/EUD Pull-Down Menu

The **PV/EUD** pull-down menu corresponds to the DESC target on the Native Window Alarm Summary. It allows the user to manually toggle between displaying the current Process Variable value or the Engineering Units description in the PV value field for all alarm lines.

Figure 15 PV/EUD Pull-Down Menu



2 GUS Alarm Summary Display

2.2 GUS Alarm Summary Header Information

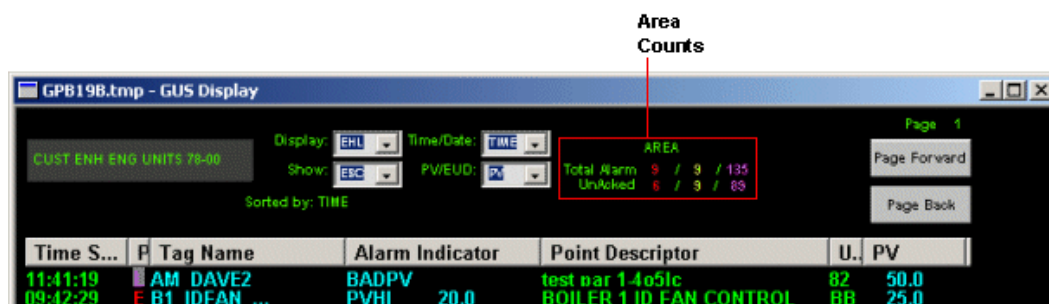
Area Counts

Alarm count information for the area appears beneath the **AREA** heading. Following the text “Total Alarm”, three sets of numbers separated by forward slashes (/) appear. The numbers, representing the total number of Emergency, High, and Low alarms for the area, correspond to the area alarm counts displayed in the Native Window Alarm Summary Display.

Beneath **Total Alarm**, the text “UnAcked” appears, followed by three sets of numbers. The numbers represents the total number of unacknowledged Emergency, High, and Low alarms in the area; this feature is not available in the standard Native Window Alarm Summaries.

In the event of an Alarm List Overflow or a locally Lost Process and/or Hiway Event, this data is overwritten with the appropriate text in reverse video cyan.

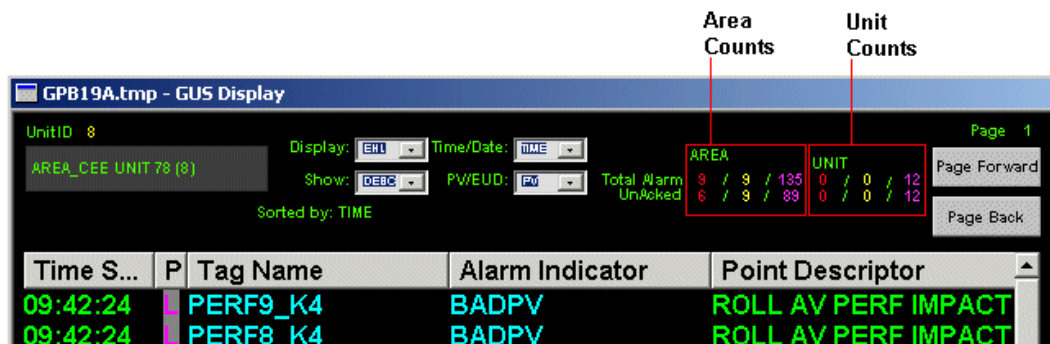
Figure 16 Area Alarm Summary – Area Counts



Unit Counts

Since the unit alarm count information is specific to a given unit, it does not appear on the GUS Area Alarm Summary. The GUS Unit Alarm Summary shows the unit count information beneath the **UNIT** text heading. Total unit alarms and total unacknowledged alarms for the unit are displayed. Unacknowledged unit counts is a feature not available in the standard Native Window Unit Alarm Summary.

Figure 17 Unit Alarm Summary – Area and Unit Counts



Collection Rate

Default Collection Rate

The first time GUS Display Builder is invoked after it is installed, an initial collection rate of 4 seconds and a group ID of 1 are specified for the display.

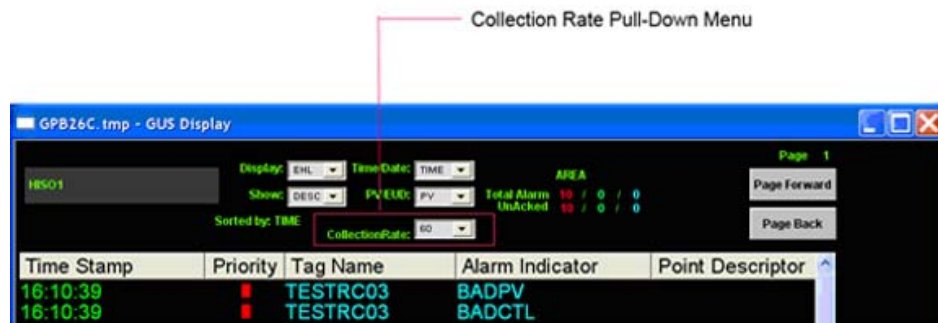
The display should be saved before validating to see the updated display name. The changes made to the Collection rate and Group ID will be reflected in the display when the display is saved. You can generate your own Collection Rate files by making copies of the Collection Rate template file and then editing the copies using either Notepad or MS Excel. A collection rate of 4 seconds and a Group ID of 1 are used if the collection rate is not specified for a parameter name in the file.

2 GUS Alarm Summary Display

2.2 GUS Alarm Summary Header Information

The Current PV for the points in alarm summary will be collected at "Collection rate". The data is collected for the Current PV if the GUS Alarm OCXs are running.

Figure 188 Collection Rate Pull-Down Menu



Page information

The current page number is displayed in the upper right-hand corner; however, the total number of pages available for viewing is not displayed as it is on the Native Window Alarm Summaries (that is, Page 1 of 4).

The page forward and page back functionality works in the same manner as the Native Window Alarm displays; that is, page wrapping is supported, and a maximum of five pages (100 alarms) can be viewed. Two selectable buttons are available on the GUS Alarm Summaries for paging forward and back. The buttons are located directly beneath the current page number and are labeled **Page Forward** and **Page Back**.

2.3 GUS Alarm Summary Alarm Line Information

Alarm Line Column Headings

Column headings appear on GUS Alarm Summary displays, but not on Native Window Alarm Summary displays. The column headings describe each field of data associated with the alarm line. The order of the fields is fixed; however, you may modify each field width by dragging the column separator. The **Time Stamp** and **Priority** column headings perform a function when selected. (Refer to the section Dynamic Sorting.)

The column heading text associated with the point descriptor field changes according to the type of data selected for display in the field. The column heading text displays either **Point Descriptor**, **PRIMMOD**, or **MPRIMMOD**. (Refer to the section Point Descriptor/Primmod/Multiple Primmod Field.)

Alarm Line Fields

The following section describes alarm line information in the order as it appears from left to right in the GUS Area and Unit Alarm Summaries.

Figure 19 Alarm Summary Alarm Line Data with Point Descriptions

Time St...	P	Tag Name	Alarm Indicator	Point Descriptor	Unit...	PV
15:06:46		52LAX8130	OFFNORM	DISC. LT8130/8030	01	ALARME
15:06:10		52LAH8130	OFFNORM	BALLON F15	01	ALARME
15:05:48		52LAH8038	OFFNORM	BALLON F13	01	ALARME
15:05:01		52ARR_MEROX2	OFFNORM	ARRET MEROX2	01	A_MEROX2
15:04:41		52FC8023	PVHI 20.0	SOLUT.SWEETENING	01	27.00
15:04:19		52FC8022	PVHI 20.0	AIR PROCEDE->F13A	01	28.00
15:04:02		52PC8021	PVHI 3.0	ASPIRATION J6	01	3.500
15:03:45		52LC8010	PVHI 75.0	BALLON F3	01	78.0
15:03:00		REG_C	OPHI 75.0	REG CONTROL FACEPLATE	01	80.0
15:02:19		52LI8034	PVHI 11.0	BALLON F5	01	89.0
15:01:57		52LI8038	PVHI 85.0	BALLON F13	01	96.0

Time Stamp Field

The **Time Stamp** field contains either the time, date, or time and date of the alarm, depending on the selection in the Time/Date pull-down menu. The field behaves identically to the Native Window Alarm Summary display in appearance and functionality.

2 GUS Alarm Summary Display

2.3 GUS Alarm Summary Alarm Line Information

Priority Field

The **Priority** field displays the priority of the alarm. Data in this field behaves identically to the Native Window Alarm Summary display in appearance and functionality. Symbols/characters and alarm color are supported, as defined in NCF configuration.

Tag Name Field

The **Tag Name** field displays the tag name of the point in alarm. It behaves identically to the Native Window Alarm Summary display in appearance and functionality.

Alarm Indicator

The **Alarm Indicator** field displays the alarm indicator, and the alarm limit for process alarms, or the Module/State for sequence program alarms. It behaves identically to the Native Window Alarm Summary display in appearance and functionality.

Point Descriptor/Primmod/Multiple Primmod Field

As with the Native Window Alarm Summary Display, the type of data appearing in the **Point Descriptor** field toggles between Point Description, Primmod, and Multiple Primmod (MPROD). Unlike the Native Window, the Multiple Primmod names appear in full size text, on one line, separated by slashes (/).

If the field width is not wide enough to accommodate the MPROD string, part of the data will be truncated and followed by an ellipsis (...). You can expand the field width by dragging the column separator. Unlike the Native Window Alarm Summary display, the **Unit ID** field is not eliminated when the MPROD setting is in effect.



ATTENTION

In the following figure, the Multiple PRIMMOD field width has been expanded to accommodate all Multiple PRIMMOD strings.

Figure 20 Alarm Summary Alarm Line Data with Multiple Primmods

Multiple Primmods & Unit IDs co-exist together

Time Sta...	Prior...	Tag Name	Alarm Indicator	MPRIMMOD	Unit...
14:34:40	E	HG1101A	PVHI 78.0		03
14:34:40	E	CBREG99	PVHI 78.0		99
14:34:40	E	CBREGSAP	PVHI 78.0		01
14:34:40	E	HG1001	PVHI 78.0		01
14:33:54	E	AMREGU7	PVHH 75.0	/THISISALONGMPROD/SHORTONE/1234567890ABCDEF/ABC/	12
14:33:27	E	AMFLAG_U7	OFF		07
	E	AMFLAG_U6	OFF		06
14:34:40	H	TIME08	PVHI 150.0		01
14:34:40	H	TIME06	PVHI 760.0		01
14:34:40	H	TIME04	PVHI 80.0		01
14:34:40	H	TIME01	PVHI 78.0		01
14:33:54	H	AMREGU7	PVHI 55.0	/THISISALONGMPROD/SHORTONE/1234567890ABCDEF/ABC/	12

17754

Unit ID Field

The **Unit ID** field behaves identically to the Native Window Alarm Summary in appearance and functionality, except **Unit ID** is not removed when you select **MPROD**.

PV Value/EU Description Field

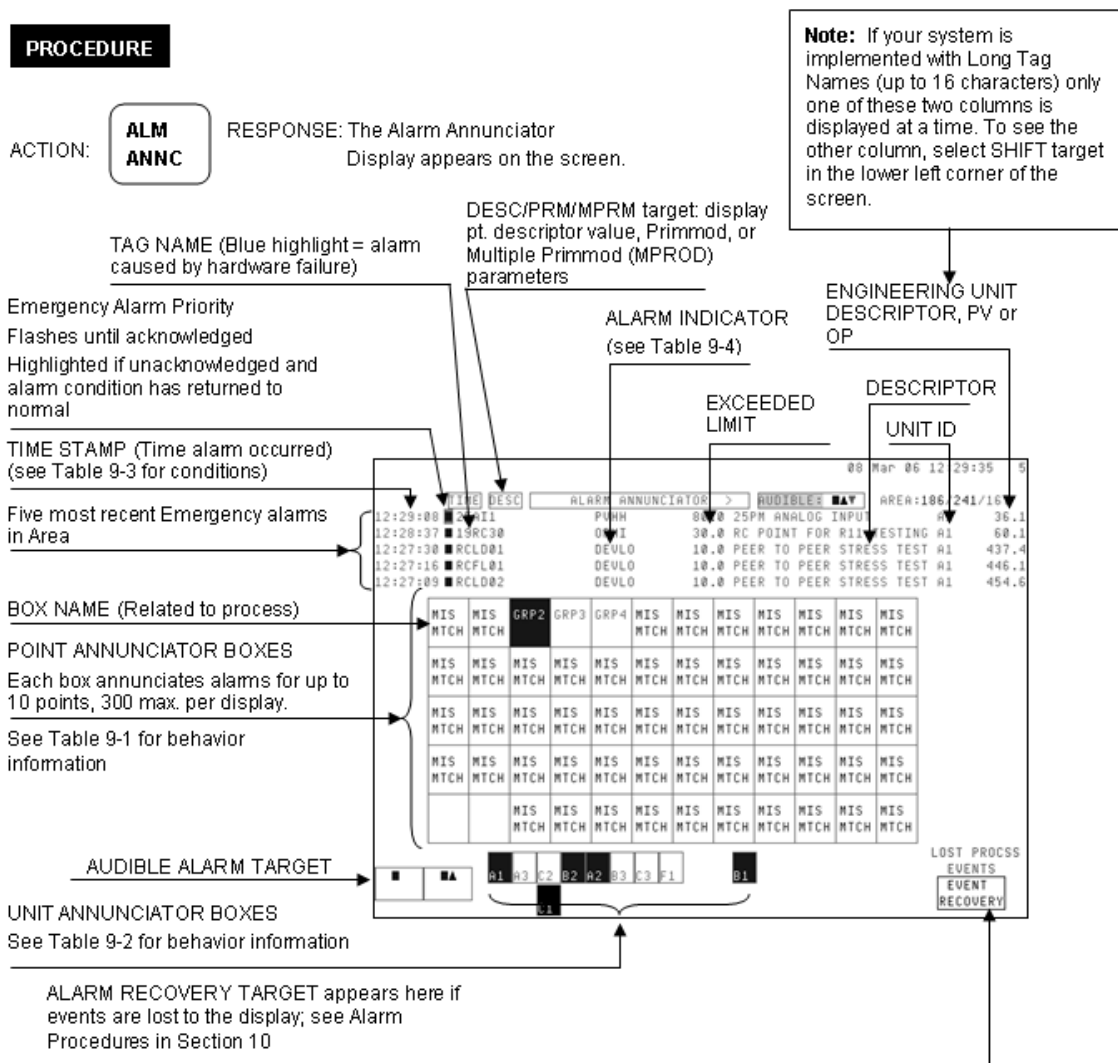
The **PV Value** field contains either the current PV Value or the Engineering Units Description of the alarm, depending on the user selection in the **PV/EUD** pull-down menu. The PV Value field behaves identically to the Native Window Alarm Summary in appearance and functionality.

With GUS R360, an option is available to display the OP values for the Output Alarms (OPHI and OPLO) instead of the PV/EUD values in the Alarm Summary Display. By default, the OP values are displayed in yellow. To select this option, changes have to be made in the NCF. Such changes have a moderate impact on the GUS. To load the revised NCF, the station must be shut down and reloaded. For more information, refer to the Network Form Instructions.

For NIM regulatory and analog output points, when the value of OUTIND is set to Reverse the reversed OP values are displayed.

2 GUS Alarm Summary Display

2.3 GUS Alarm Summary Alarm Line Information



2.4 Operating the GUS Alarm Summary Display

Alarm window

The first page of the GUS Alarm Summary does not contain a “new alarm” window like the Native Window Area and Unit Alarm Summaries; instead, a maximum of twenty alarm lines is displayed on the first page and on the remaining four pages.

Selecting an Alarm

When you select an alarm, the following occurs:

- The alarm line is highlighted in blue.
- Text appears in the upper left-hand area of the display just above the **Time Stamp** and **Priority** column headings. The text reads: “Selected Point nnnnnnnn”, where n is the point name.
- Three buttons appear in the lower left-hand corner of the display: **Disable Alarm**, **Inhibit Alarm**, and **Point Detail**.

The Disable Alarm and Inhibit Alarm buttons operate the same as in the Native Window Alarm Summaries. After button selection, a dialog box appears giving you the opportunity to confirm or cancel the action. The text in the pop-up dialog box reads: “select OK to disable alarm” or “select OK to inhibit alarm.”

Selecting **Point Detail** causes the Native Window to gain focus, displaying the DETAIL of the selected point.

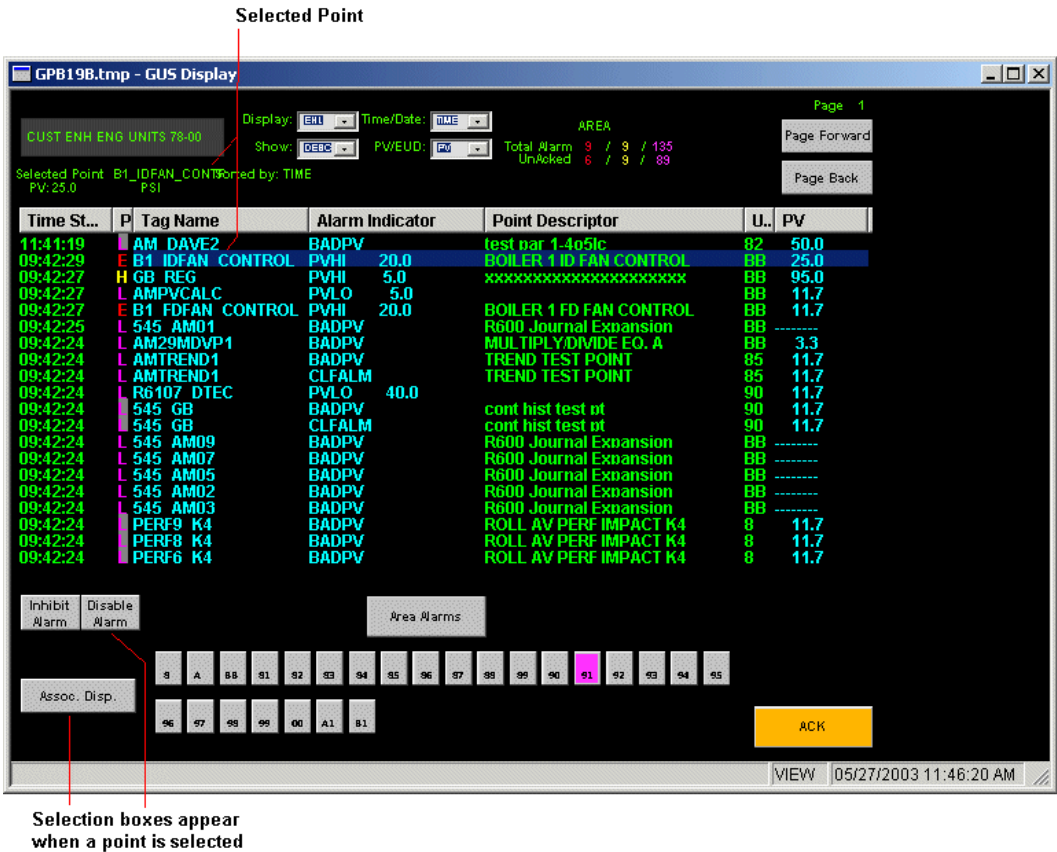
Selecting the **GUSPro Menu** button invokes the GUSPro main menu, if you have GUSPro installed on the machine.

Group key behavior

The GUS Alarm Summary does not function the same way as the Native Window Alarm Summary with regard to the Group key. If you select a point in the alarm list on the GUS Alarm Summary display and then press the Group key on the keyboard, you will get a message asking you to enter the Group Number. Executing the same actions with the Native Window Alarm Summary display results in display of the Group containing the selected point automatically, without having to enter a Group Number.

2 GUS Alarm Summary Display
2.4 Operating the GUS Alarm Summary Display

Figure 21 Selecting an Alarm from the Alarm Summary



Area Alarms button

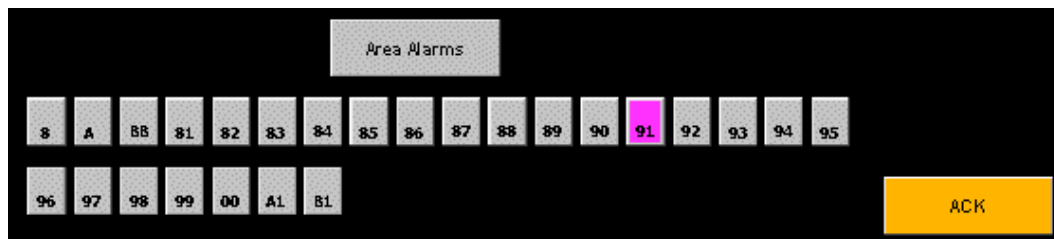
The purpose of the **Area Alarms** button, located directly beneath the alarm line data, is to navigate back to the Area Alarm Summary display from one of the Unit Alarm Summaries. When invoking the GUS Alarm Summary for the first time, the display defaults to the Area view.

Select the **Area Alarms** button from one of the Unit Alarm displays, and the Alarm Summary will change to the Area view. The button exists on both the Area and Unit Alarm Summary displays; however, selecting the button from the Area Alarm Summary has no effect upon the display. The Native Window Alarm Summary display does not have a button corresponding to this function.

Unit boxes

The Unit boxes show alarm status of all units in the area and serve as targets to call up the Unit Alarm Summary Displays. These boxes behave identically to the Native Window Alarm Summary display in appearance and functionality.

Figure 22 Area Alarms Button and Unit Boxes



Acknowledging alarms

The **ACK** button is located in the lower right corner of the GUS Alarm Summary displays. It performs the same function as the **ACK** key on the keyboard. In contrast to the Native Window Alarm Summary, *if you select a specific alarm on the GUS Alarm Summary, the **ACK** button acknowledges only that alarm; the other alarms on the page remain unacknowledged.* The **ACK** button does not exist on the Native Window Alarm Summary.

Area Change

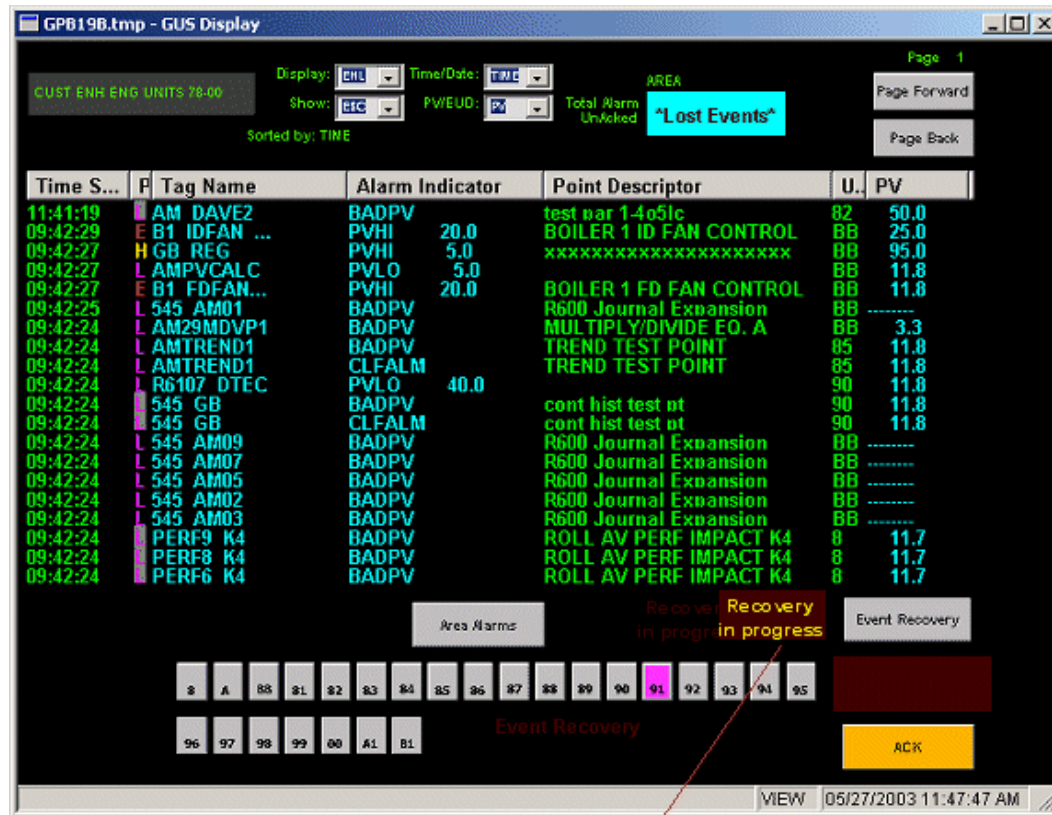
Unlike the TPS Network, when an Area Change is in progress, the GUS Area or Unit Alarm Summary remains visible; however, the data associated with the Area or Unit is not available. When the Area Change is complete, the GUS displays show alarm data associated with the new area.

2.5 GUS Lost Event Indication and Recovery

For lost Data Hiway and process events, their messages, indicators, and methods of retrieval are handled the same in the GUS and Native Window Alarm Summary displays.

The lost event and recovery targets are located in the lower right-hand corner of the GUS Alarm Summary display, just above the ACK button.

Figure 23 Area Alarm Summary - Locally Lost Events with Recovery in Progress



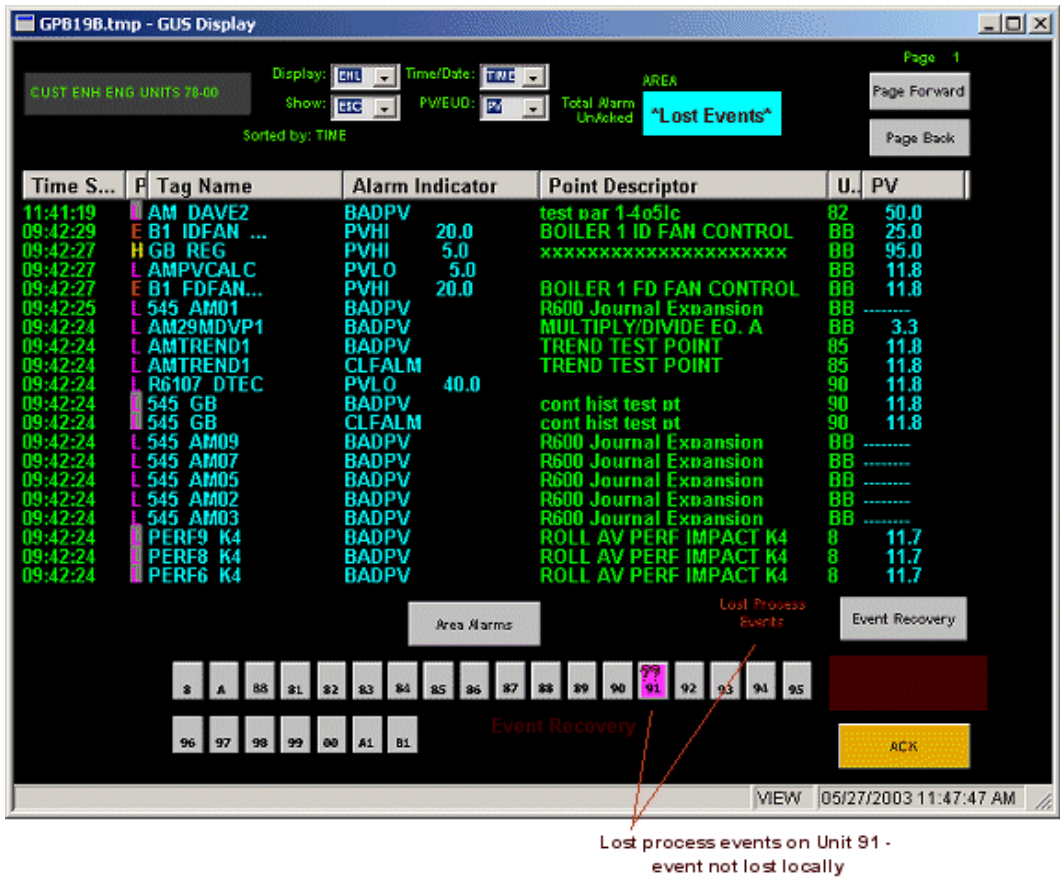
Locally lost events with
Recovery in progress



ATTENTION

The event in the following example was not lost locally

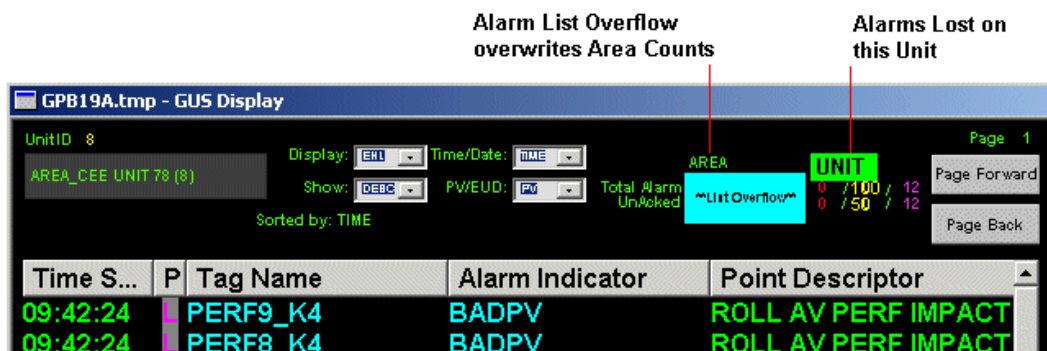
Figure 24 Area Alarm Summary - Lost Process Events on Unit 91



2.6 GUS Alarm List Overflow

The standard TPS Network method of processing and retrieving points caused by an alarm list overflow applies to the GUS Alarm Summaries.

Figure 25 Alarm List Overflow - Unit Alarm Summary of the Affected Unit



3. GUS Faceplate Control

3.1 Faceplate Control Overview

Functional description

The GUS Faceplate Control (OCX) is an OLE Control for viewing and manipulating the parameters of a selected point.

Once configured, the Faceplate Control provides the following access:

- Access to process variable, mode, and mode attribute parameters using both the standard Windows GUI components and the Integrated keyboard (IKB) or Operator Entry Panel (OEP).
- Access to the Native Window detail point or associated display for the specific point.

The Faceplate Control mimics the functionality of one of the point positions in a Native Window (R6xx) group display.

In addition to GUS displays, any OCX container application can use the Faceplate Control.



REFERENCE - INTERNAL

For details on implementing an equivalent faceplate control (the TPS Faceplate Control) in HMI web displays for ES-T nodes, refer to the *Integrated Experion-TPS Nodes (ES-T, ESVT, & ACE-T) User's Guide, EX23*.

Runtime and Buildtime modes

The Faceplate Control has a buildtime and a runtime mode.

The buildtime mode provides a property page to allow you to configure the point to be displayed at runtime. The buildtime mode display is the same for all point types. It consists of a dark gray rectangle, with a light gray rectangle bordering the upper third.

The buildtime mode display, with the addition of a red “X” bisecting the rectangle, is the initial display that appears at runtime. The red “X” indicates that the Faceplate has not completed the LXS data access connections. The Faceplate is not fully operational until the red “X” disappears.

The runtime mode display is specific to each point type. The Faceplate selects the combination of graphical elements to represent a point type, based on the point's display type parameter.

3 GUS Faceplate Control

3.1 Faceplate Control Overview

Faceplate resizing

At buildtime, after inserting the Faceplate control into a GUS display, you must drag the object (define the height and width) to display the entire Faceplate. The Faceplate Control does not supply a default size.

If the display containing the Faceplate Control is re-sizeable at runtime, the font size can become too small to read. You are responsible for choosing a size large enough to be readable at runtime.

The Faceplate Control in the GUS Group Display and the Faceplate EXE maintain the original size of 125 pixels wide and 550 pixels high.

Supported Data Point types

The following table lists the point types supported by the GUS Faceplate Control.

Table 3-1 Supported Data Point Types

Supported Point Type	Point Types
AM Points	REGAM, FLAGAM, NUMERCAM, TIMERAM, COUNTAM, SWITCHAM, CUSTOMAM
NIM Points	ANINNIM, ANOUTNIM, DIINNIM, DIOUTNIM, DICMPNIM, REGPVNIM, REGCLNIM., LOGICNIM, FLAGNIM, TIMERNIM, NUMERNIM, DEVCTL, PRMODNIM
HG Points	ANLINHG, ANLOUTHG, ANLCMPHG, DIGINHG, DIGOUTHG, DIGCMPHG, REGHG, COUNTHG, HGTIMER, FLAGHG, PRCMODHG, CTLCOUNT, NUMERCHG, LGCBLKHG
Fieldbus Points (CM Point Types)	FBCM (ANALGIN, ANALOUT, PID) PECM and SECM do not show Control Data

Supported IKB and OEP keys

The following table lists the keys on the IKB and OEP that the GUS Faceplate Control supports for points.

Table 3-2 Supported IKB and OEP Keys

Raise	NORM
Lower	AUTO
Fast raise	MAN
Fast Lower	Detail
SP	Enter
OP	Numeric Keys <0 – 9>
CLR ENTR	Numeric Keypad *

Faceplate constraints

The GUS Faceplate Control has the following constraints:

- The faceplate is not resizable in buildtime or runtime; that is, you cannot make it bigger or smaller by using the zoom feature or by setting the display property as zoom-to-fit. The Faceplate control does not zoom-to-fit.
- The colors and text of the Faceplate are not user configurable through the property page or scripting.
- Only four Faceplates may be running on a single GUS node. If the GUS Group Display is running, then only three faceplates may run.
- When a point.parameter with Engineering keylevel access is changed through some other display, changes to parameters caused by the Engineering change will not be displayed in the Faceplate until it is re-invoked for that point. You must invoke the Faceplate for a point to see changes made under this condition
- The Faceplate only displays and manipulates points on the TPN that are accessed through the LXS. It cannot display points from an HCI server.

3.2 Faceplates in Runtime Mode

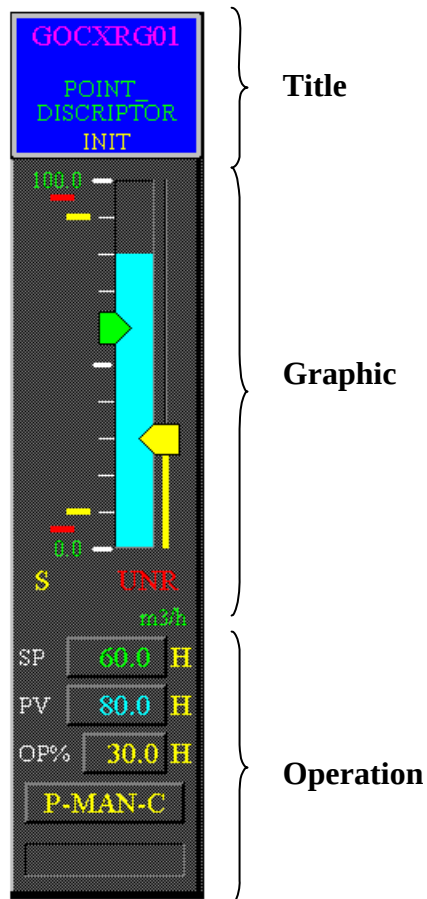
Major parts of Faceplates

The GUS runtime Faceplate Control has the following parts:

- Title area - Displays the point name, point descriptor, and the point status.
- Graphic area – Graphically displays the value of the PV, OP, and SP. The content varies based on the display type. If available, it displays system alarms, alarm enable status, alarm status, and engineering unit description. Digital point types show state boxes, descriptions, and choices. Regulatory types display grids, trip points, and a ramping bar. This area also displays the mode or state choices available for user modification.
- Operation area - Displays the current values and enables you to select and manipulate the values. It may consist of one or more buttons representing SP, PV, OP, Mode, OK, and a data entry box.

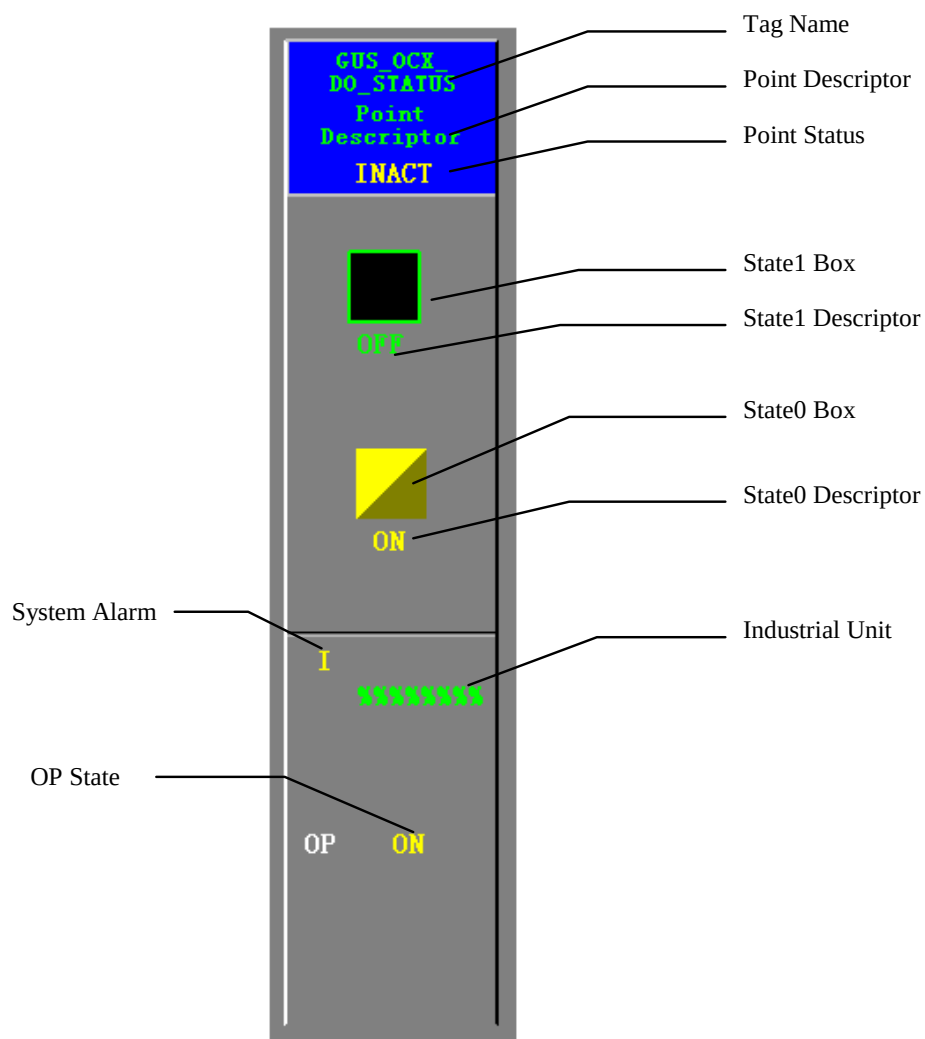
Regulatory Control Faceplate Example

Figure 26 Regulatory Control Faceplate at Runtime



Digital Output Faceplate Example

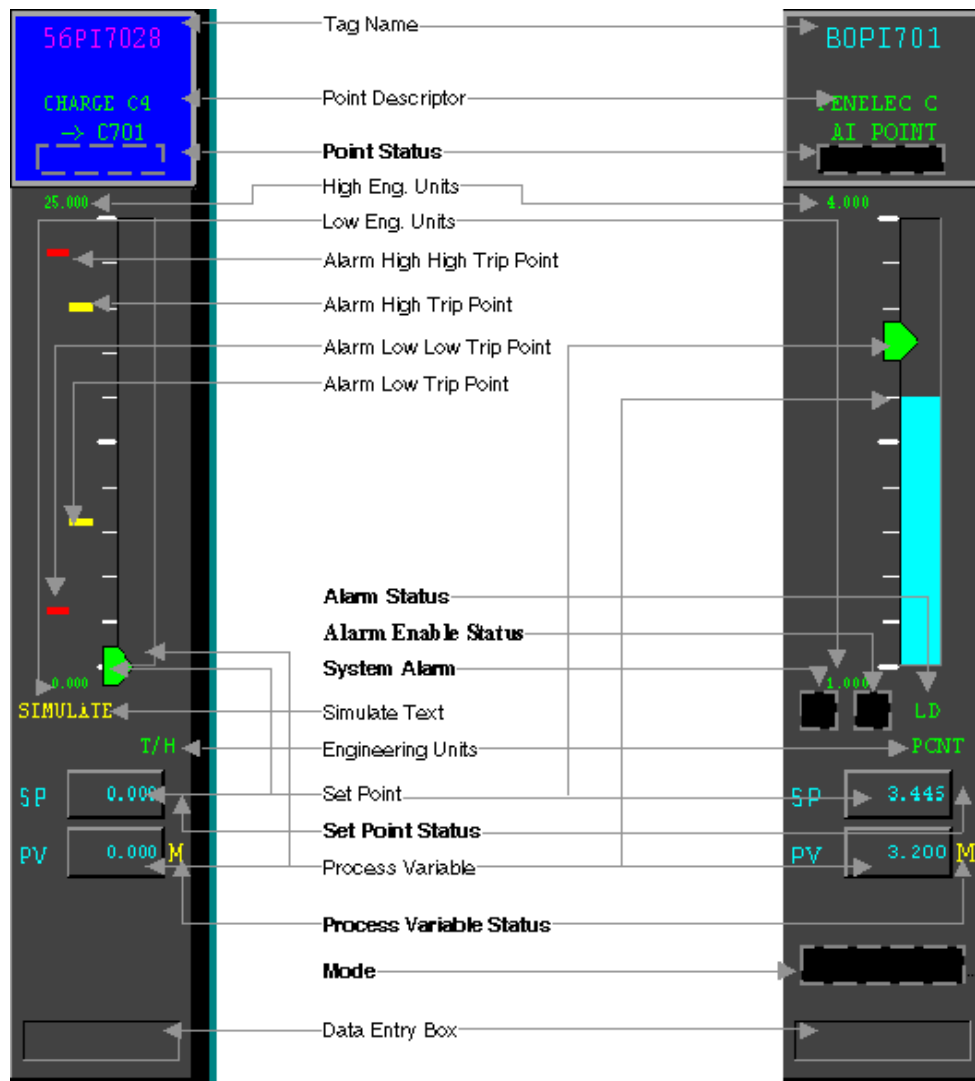
Figure 27 Digital Output Point at Runtime



Faceplate definition - Analog Input Points

Dashed boxes in the following examples indicate parameters unavailable in the example.
Labels in bold have additional information described in tables following the example.

Figure 28 Analog Input Faceplate Displays



3 GUS Faceplate Control

3.2 Faceplates in Runtime Mode

Analog Input Alarm Status

The label **Alarm Status** on Figure 28 Analog Input Faceplate Displays identifies the area where the Alarm Status indicator appears on the Faceplate.

Table 3-3 AI and PID Alarm Status Indicators

Indicator	Meaning
[BLANK]	Point not in alarm or status is not accessible.
(Yellow)*	Overview Tracer; indicates when point's deviation from the setpoint is greater than the overview value.
ADV	Advisory Deviation
BP	Bad Process Variable
HD	High Deviation
2HD	High High Deviation
HP	High Process Variable
2HP	High High PV
INH	Alarms are inhibited.
LD	Low Deviation
LP	Low Process Variable
2LP	Low Low PV
OFN	Off Normal
ORC	Output Rate of Change
PSC	PV Significant Change
ROC	Rate of Change
RC+	Positive Rate of Change
RC-	Negative Rate of Change
Note*: Indicator color (red or yellow) corresponds to alarm's priority. For example: Red = Emergency, Yellow = High. This is defined in the system wide values.	

Analog Input Point Status

The label **Point Status** on Figure 28 Analog Input Faceplate Displays identifies the area where the Point Status indicator appears on the Faceplate.

Table 3-4 AI Point Status Indicators

Indicator	Meaning
[BLANK]	Point is in Active state
INACTIVE	Point is in Inactive State

Analog Input Alarm Enable Status

The label **Alarm Enable Status** on Figure 28 Analog Input Faceplate Displays identifies the area where the Alarm Enable Status indicator appears on the Faceplate.

Table 3-5 AI and PID Alarm Enable Status Indicators

Indicator	Meaning
[BLANK]	Alarm is Enabled (inhibited alarms are displayed in the Alarm Status area).
DIS	Alarm status is Disabled.
INH	Alarm status is Inhibited.

3 GUS Faceplate Control

3.2 Faceplates in Runtime Mode

Analog Input System Alarm

The label **System Alarm** on Figure 28 Analog Input Faceplate Displays identifies the area where the System Alarm indicator appears on the Faceplate.

Table 3-6 AI, AO, and PID System Alarm Indicators

Indicator	Meaning
[BLANK]	Point is in OK state.
F (Yellow)	Point's board is in device failed. (PIU, MC, PM, APM, LM).
I (Yellow)	Point is in IDLE state (MC, PM, APM, LM).
R (Yellow)	Point is in RESET state.
S (Yellow)	Point's backup box is in operation (data saved to the backup controller).
U (Yellow)	DHP not scanning point; possibly a bad PC-configuration address.

Analog Input SP Status

The Set Point Status indicators appear to the right of the SP value. The label **Set Point Status** on Figure 28 Analog Input Faceplate Displays identifies this area. Indicators in bold are Fieldbus values.

Table 3-7 AI SP Status Indicators

Indicator	Meaning
[BLANK]	Normal or No data
B (Red)	Bad Setpoint
H (Yellow)	Setpoint High Limit
H (White)	Setpoint High Limit (FF device)
L (Yellow)	Setpoint Low Limit
L (White)	Setpoint Low Limit (FF device)
C (White)	Setpoint Constant (FF device)

Analog Input Mode

The Mode indicators appear directly below the PV value. The label **Mode** on Figure 28 Analog Input Faceplate Displays identifies this area. Fieldbus Analog Input Points are the only Analog Input Points that have a Mode.

Table 3-8 AI Mode Indicators

Indicator	Meaning
AUTO	AUTOMATIC: Calculated by algorithm based on Operator setting.
MAN	MANual: Controlled by Operator.
OOS	Out of Service
???	Mode is inaccessible
!!!	Mode is indeterminate
---	Mode is invalid

3 GUS Faceplate Control

3.2 Faceplates in Runtime Mode

Analog Input PV Status

The Process Variable Status indicators appear to the right of the PV value. The label **Process Variable Status** on Figure 28 Analog Input Faceplate Displays identifies this area. Indicators in bold are Fieldbus-related values.

Table 3-9 AI PV Status Indicators

Indicator	Meaning
[Blank]	Normal
B (Red)	Bad PV
H (Red)	Bad PV and PV Hi Extended Range Violation Bad PV and PV High Limit Indication (FF device)
L (Red)	Bad PV and PV Low Extended Range Violation Bad PV and PV Low Limit Indication (FF device)
C (Red)	Bad PV and Constant Indication (FF device)
M (Red)	Bad PV and PVSOURCE is Manual
S (Red)	Bad PV and PVSOURCE is Substituted
U (Yellow)	PV is Uncertain
H (Yellow)	PV is Uncertain and PV Clamped to Hi Extended Range PV is Uncertain and PV Clamped to Hi Limit Indication (FF device)
L (Yellow)	PV is Uncertain and PV Clamped to Lo Extended Range PV is Uncertain and PV Clamped to Lo Limit Indication (FF device)
M (Yellow)	PV is Uncertain and PVSOURCE is MANual
S (Yellow)	PV is Uncertain and PVSOURCE is Substituted PV is Uncertain and Substituted (FF device)
C (Yellow)	PV is Uncertain and Constant (FF device)
H (White)	PV is Good and Hi Limit Indication (FF device)
L (White)	PV is Good and Lo Limit Indication (FF device)
C (White)	PV is Good and Constant (FF device)

Faceplate definition - Analog Output Points

The following figure shows Analog Output Faceplate examples. Dashed boxes indicate the location for graphic elements unavailable on the example displays. Labels appearing in bold have additional information described in the tables following the example.

Analog Output Point Status

The label **Point Status** on Figure 28 Analog Input Faceplate Displays identifies the area where the Point Status indicators appear on the Faceplate.

Table 3-10 AO Point Status Indicators

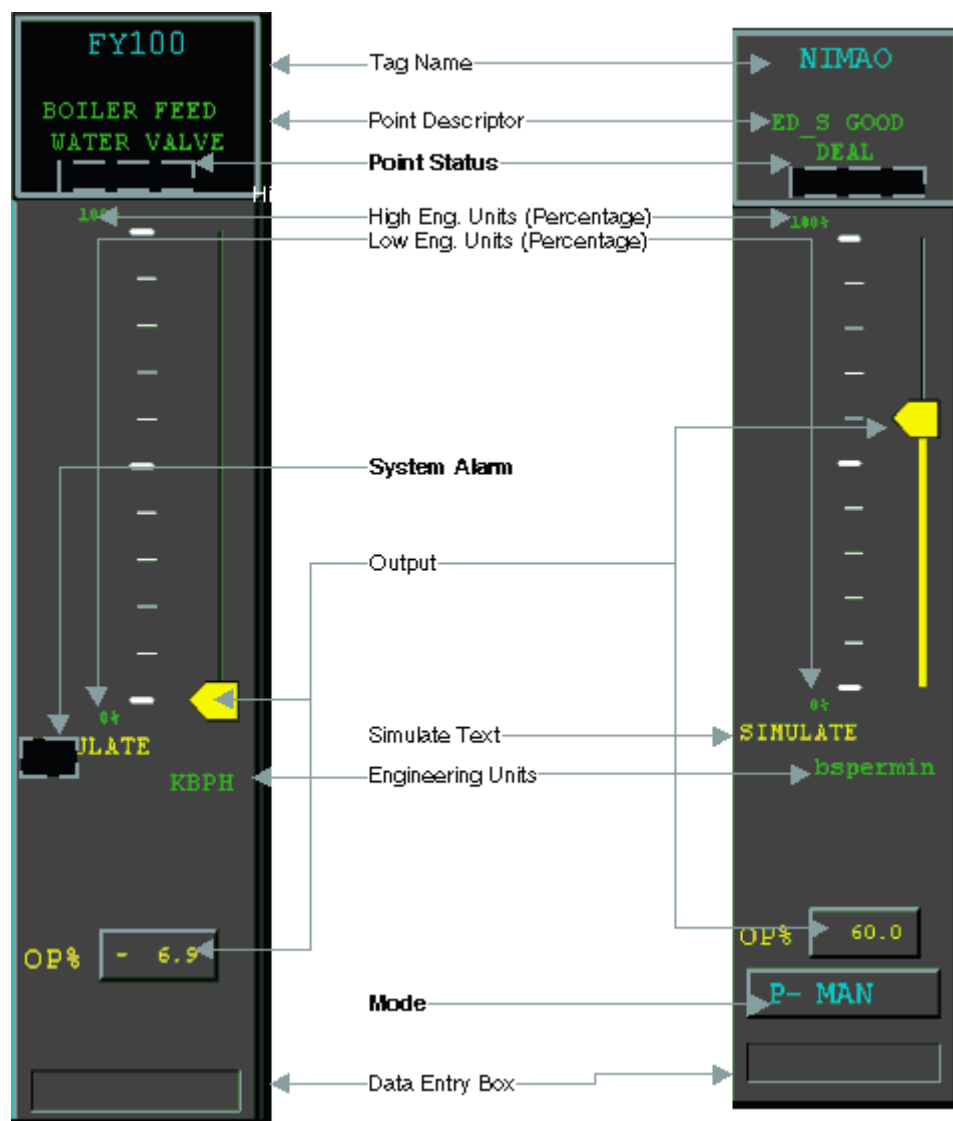
Indicator	Meaning
[BLANK]	Point is in active state.
INACTIVE	Point is in inactive state.
RED TAG	Point is red tagged.
STDBYMAN	Point is in stand-by manual mode.

Analog Output System Alarm

The label **System Alarm** on Figure 28 Analog Input Faceplate Displays identifies the area where the System Alarm Status indicators appear on the Faceplate.

Refer to the Table 3-6 AI, AO, and PID System Alarm Indicators for more details.

Figure 29 Analog Output Faceplate Displays



Analog Output Mode

The Mode indicators appear directly below the OP value. The label **Mode** on Figure 28 Analog Input Faceplate Displays identifies this area. Non-Fieldbus TPS Points have only MAN, AUTO, CAS, BCAS, and NORM as mode selections. Selections vary based on the point type.

Table 3-11 AO Mode Indicators

Indicator	Meaning
AUTO	AUTOMATIC: OP is calculated by algorithm based on Operator setting.
BCAS	Backup Cascade: OP receives its setpoint from the OP of a primary data point
CAS	CASCADE: OP is from an AM or a CM, or from a point in the same Hiway controller file.
IMAN	Initialization MANUAL
LO	Local Override
MAN	MANUAL: OP is controlled by the Operator.
NORM	Normal Mode
OOS	Out of Service
RCAS	Remote Cascade
ROUT	Remote Output (applicable to PID)
???	Mode is inaccessible
!!!	Mode is indeterminate
---	Mode is invalid

3 GUS Faceplate Control

3.2 Faceplates in Runtime Mode

Faceplate definition - Regulatory Control Points

The following figure shows a Regulatory Control (PID) Faceplate. Dashed boxes indicate the location for graphic elements unavailable on the example display. Labels appearing in bold have additional information described in the tables following the figure.

PID Point Status

The label **Point Status** in the figure identifies the area where the Point Status indicators appear on the Faceplate.

Table 3-12 PID Point Status Indicators

Indicator	Meaning
[BLANK]	Point is in Active state.
INACTIVE	Point is in Inactive state.
STDBYMAN	Point is in stand-by Manual mode.
INIT	Point is being initialized.
OR in SEL	Point selected by override selector.
OR Not SEL	Point not selected by override selector.
RED TAG	Point is red tagged.
Custom String(8)	User configured status message for RED TAG state.

PID Alarm Status

The label **Alarm Status** in the figure identifies the area where the Alarm Enable Status indicators appear in the Faceplate.

Refer to Table 3-3 AI and PID Alarm Status Indicators for a list of all the indicators and what they mean.

PID System Alarm Status

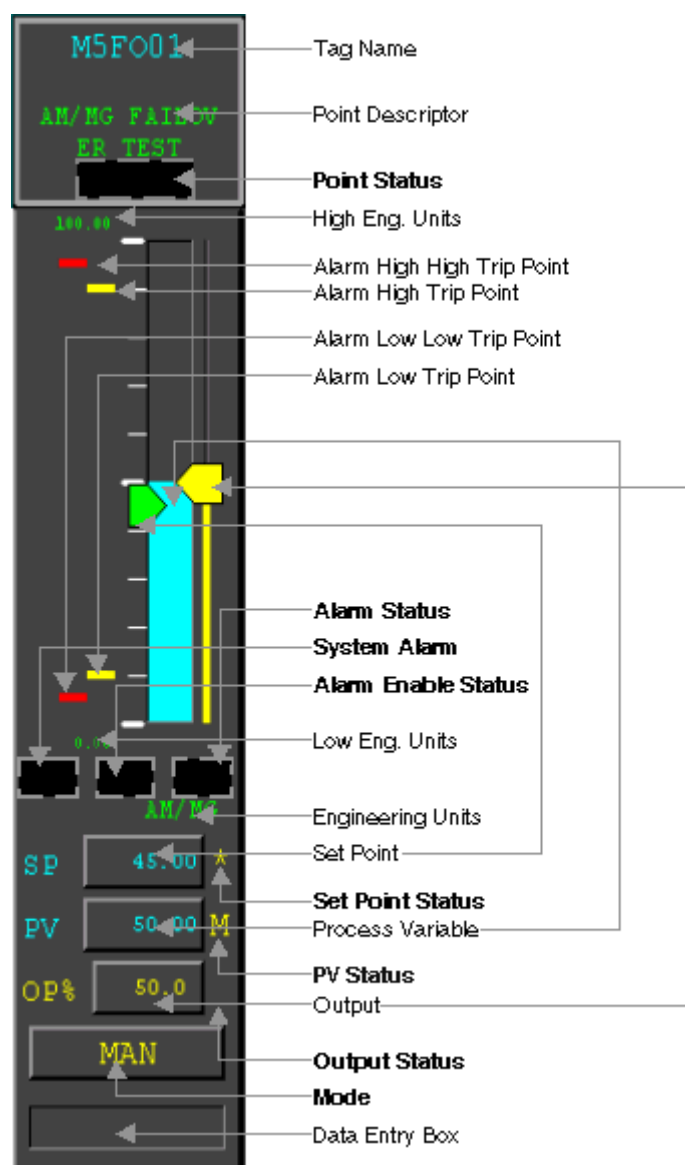
The label **System Alarm** in the figure identifies the area where the System Alarm indicators appear in the Faceplate..

Refer to Table 3-6 AI, AO, and PID System Alarm Indicators for a list of all the indicators and what they mean.

3 GUS Faceplate Control

3.2 Faceplates in Runtime Mode

Figure 30 PID Faceplate Display



PID Alarm Enable Status

The label **Alarm Enable Status** in the figure identifies the area where the Alarm Enable Status indicators appear in the Faceplate.

Refer to Table 3-3 AI and PID Alarm Status Indicators for a list of all the indicators and what they mean.

PID SP Status

The Set Point Status indicators appear to the right of the SP value. The label **Set Point Status** on identifies this area. Indicators in bold are new/extended Fieldbus values.

Table 3-13 PID SP Status Indicators

Indicator	Meaning
[BLANK]	Normal or No data
B (Red)	Bad Setpoint
H (Yellow)	Setpoint High Limit
H (White)	Setpoint High Limit (FF device)
L (Yellow)	Setpoint Low Limit
L (White)	Setpoint Low Limit (FF device)
C (White)	Setpoint Constant (FF device)
P	Target Value Processor State Preset : AM only
R	Target Value Processor State Run: Am only
*	Target Value Processor State Off No control action occurs when point is in Manual mode. NOTE: Setpoint windup HiLo does not apply to FF devices.
^	Setpoint cannot be increased because of output limiting (TPS windup high).
v	Setpoint cannot be decreased because of output limiting (TPS windup low).

3 GUS Faceplate Control

3.2 Faceplates in Runtime Mode

PID Output Status

The Output Status indicators appear to the right of the OP value. The values in bold are new/extended Faceplate values.

Table 3-14 PID Output Status Indicators

Indicator	Meaning
[BLANK]	Normal
* (Yellow)	Secondary of this point is not in Cascade mode (TPS points)
* (Yellow)	Secondary of this point is not accepting Cascade mode (FF device)
* (Red)	Secondary of this point is in fault active state (FF device)
* (Green)	Secondary of this point is requesting initialization (FF device)
B (Red)	Bad output (FF device only)
H (Red)	Bad output and Hi Limit indication (FF device)
L (Red)	Bad output and Lo Limit indication (FF device)
C (Red)	Bad output and Constant indication (FF device)
F (Red)	Initiate Fault-State indication to downstream block (FF device only)
H (Yellow)	Output is at the OP High Limit (TPS and FF device)
L (Yellow)	Output is at the OP Low Limit (TPS and FF device)
^ (Yellow)	Secondary of this point is at Setpoint or Output High Limit, or reset High action limited
v (Yellow)	Secondary of this point is at Setpoint or Output Low Limit, or reset Low action limited
C (Yellow)	Output Constant indication (FF device)
A (Green)	Initialization Acknowledgement indication to downstream block (FF device only)

PID PV Status

The PV Status indicators appear to the right of the PV value. The label **PV Status** on identifies this area. The values in bold are new/extended Faceplate values.

Refer to Table 3-9 AI PV Status Indicators for a list of all the indicators and what they mean.

PID Mode

The Mode indicators appear directly below the Output value. The label **Mode** on identifies this area. The values in bold are new/extended Faceplate values.

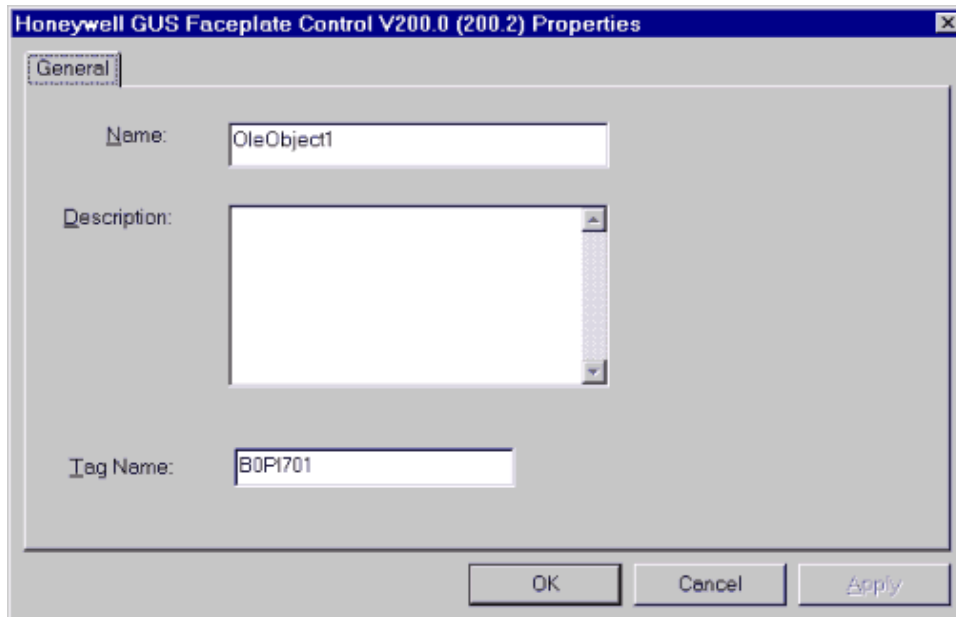
Refer to Table 3-11 AO Mode Indicators for a list of all the indicators and what they mean.

3.3 GUS Faceplate Property Pages

Description

There is a single GUS Faceplate Property Page. Here is an example of a typical Property page.

Figure 31 GUS Faceplate Property Page



There are three fields on the Property page.

- **Name** - Runtime read-only property that provides the Name for the control.
- **Description** - Runtime read-only property that provides the documentation Description for the control
- **Tag Name** - Property that provides the tag name identification for the point to be displayed

Setting Faceplate Properties

Step	Action
1	Select the Faceplate Control to be modified. RESULT: Handles at the corners and midpoint edges identify an item as selected.
2	From the Edit Menu, select Object Properties or, alternatively, select Properties from the right mouse button menu. (Double-clicking on the control also displays the Property Sheet.) RESULT: The Control displays its Property sheet. The General Tab is the default tab displayed, indicated by the highlighting rectangular box surrounding the word General.  ATTENTION: Initially, the Name options on the tab will have a default value. Subsequently, any value applied and saved for a property will appear as the default value.
3	Accept the default value for the Name or change it, as desired. – Access Name by clicking on the Name edit box, using the Tab key to navigate to the edit box, or using the Alt key and the accelerator N.  ATTENTION: A legal name must begin with an alphabetic character, and can have up to 23 additional alphanumeric or underscore characters.
4	Accept the value for the Description or change as desired. – Access Description by clicking on the Description edit box, using the Tab key to navigate to the edit box, or using the Alt key and the accelerator D.  ATTENTION: The length limit for Description is 1000 characters.
5	Accept the value for Tag Name or change as desired. – Access Tag Name by clicking on the Tag Name edit box, using the Tab key to navigate to the edit box, or using the Alt key and the accelerator T.

3 GUS Faceplate Control

3.4 GUS Faceplate Scripting Properties and Methods

3.4 GUS Faceplate Scripting Properties and Methods

GUS Faceplate Properties

The following table describes GUS Faceplate properties.

Table 3-15 GUS Faceplate Properties

Property	Description	Default Value	Script Type (C++ Type)	Configurable Through Property Page	Scripting Access
Description	Runtime read-only property that provides the documentation Description for the control	" "	String(1000)	Yes	Read Only
Name	Runtime read-only property that provides the Name for the control.	OleObject1	String(24)	Yes	Read Only
Tag Name	Property that provides the tag name identification for the point to be displayed.	" "	String(19)	Yes	Read /Write
Visible	Extended OLE Control property that provides the indicator for visibility	TRUE	Boolean	No	Read /Write

GUS Faceplate Methods

The following table describes GUS Faceplate methods.

Table 3-16 Faceplate Methods

Method	Definition	Parameter	Return
OnError	Method used to receive a string error message	String	None
SetFocus	Method used to set the focus to the Faceplate	None	None

GUS Faceplate Events

The following table describes GUS Faceplate events.

Table 3-17 GUS Faceplate Events

Faceplate Event	Definition	Parameter
ButtonClicked	An event automatically executed at runtime, when a button is clicked on the faceplate.	Integer Button 1 = PV clicked 2 = SP clicked 3 = OP clicked 4 = MODE clicked 5 = OK clicked 6 = POK clicked 7 = MAN clicked 8 = AUTO clicked 9 = CAS clicked 10 = BCAS clicked 11 = PROG/OPR clicked 12 = RCAS clicked 13 = ROUT clicked 14 = OOS clicked
OnError	An event automatically executed at runtime, when an error occurs when changing a point.parameter on the faceplate. Refer to the topic OnError Event for details.	String Message This is the error message.

3 GUS Faceplate Control

3.4 GUS Faceplate Scripting Properties and Methods

OnError Event



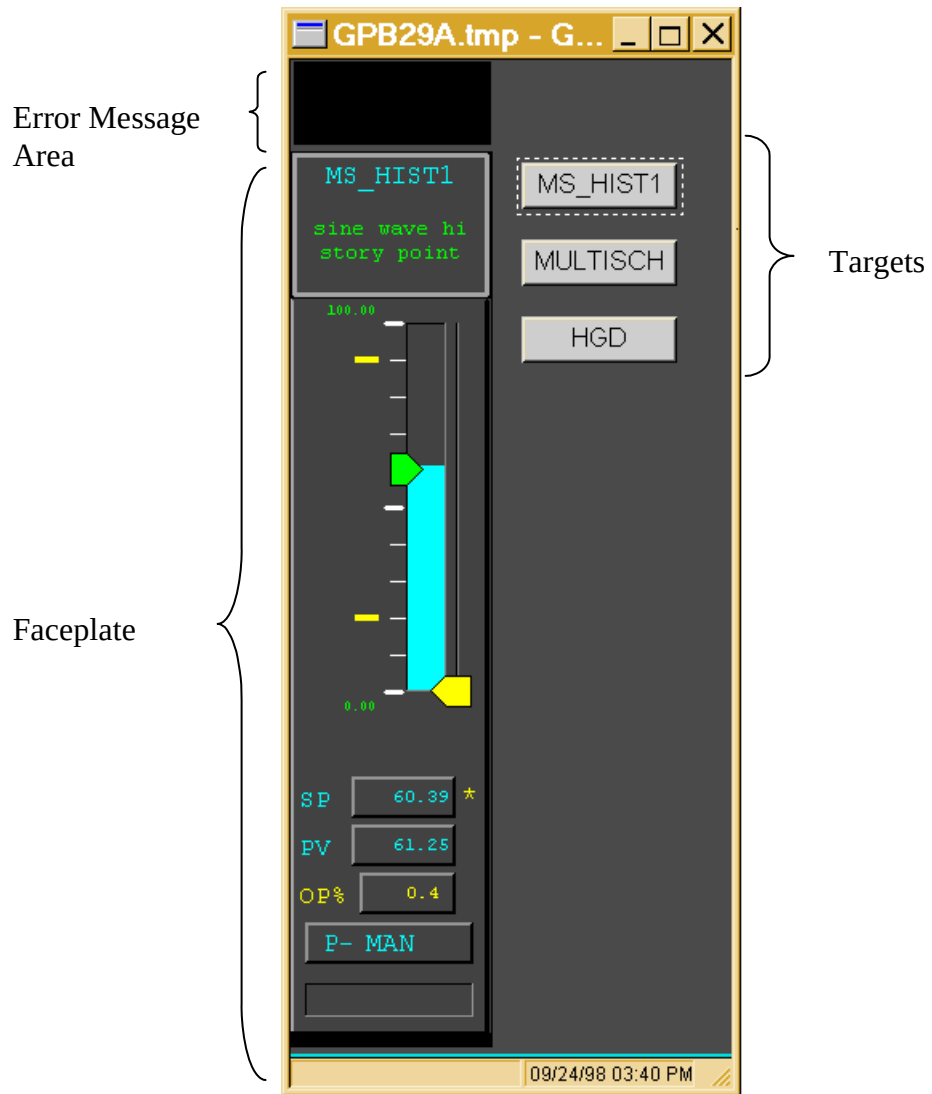
ATTENTION

There is no default error handling for the GUS Faceplate Control.

You must script the OnError event to display the error. The error message is a parameter of the OnError event. Consider using a text object or a message box to display the message to the user.

The example display in the following figure shows the error message to the user in a text object and then clears the error message text when a different button is clicked.

Figure 32 Text Object Error Message

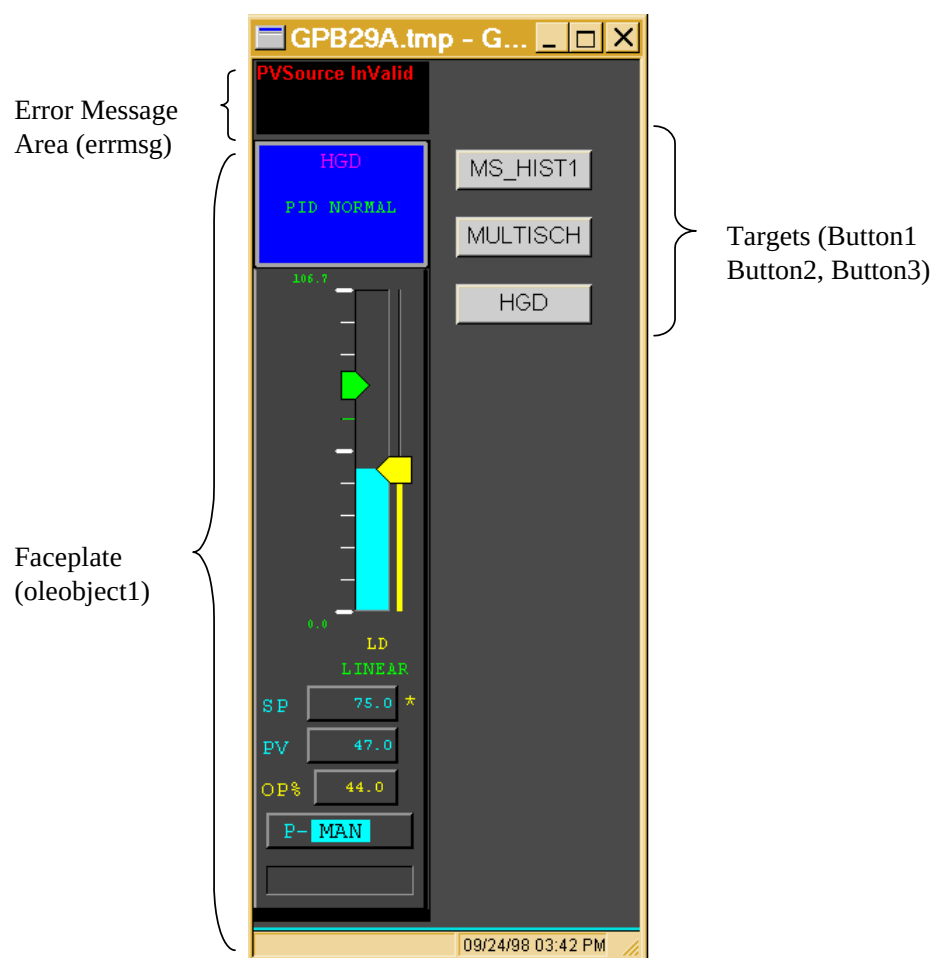


3 GUS Faceplate Control

3.4 GUS Faceplate Scripting Properties and Methods

The following figure shows the same display as in the previous figure, but with script on the OnError event, displaying a message in the errmsg text object.

Figure 33 ERRMSG Text Object Error Message



GUS Faceplate Scripting Examples

Script on Faceplate Control, oleobject1.

```
Sub ButtonClicked(Button As Integer)
    errmsg.text = ""
    'clears out the errmsg object when
    a different button is clicked
End Sub
Sub OnError(Message As String)
    errmsg.text = message
End Sub
```

Script on Targets

Using a DispDB Object

```
Sub OnRButtonClick()
    OleObject1.TagName = dispdb.[$cz_enty].[name]
End Sub
```

Button1

```
Sub OnLButtonClick()
    OleObject1.TagName = "ms_hist1"
End Sub
```

Button2

```
Sub OnLButtonClick()
    OleObject1.TagName = "multisch"
End Sub
```

Button3

```
Sub OnLButtonClick()
    OleObject1.TagName = "hgd"
End Sub
```

3 GUS Faceplate Control

3.4 GUS Faceplate Scripting Properties and Methods

Read a Point Name From an Entry Box

```
Sub OnLButtonClick()  
    OleObject1.TagName = AskBox$ ("Enter Point Name >")  
End Sub
```

Set the Name and Then Set Focus to the Faceplate

```
Sub OnLButtonClick()  
    If Faceplate.tagname = "M5F001" Then  
        Faceplate.tagname = "M5F002"  
    Else  
        Faceplate.tagname = "M5F001"  
    End If  
    Faceplate.SetFocus  
End Sub
```

Invoking a Faceplate for a Point

A Faceplate for a Fieldbus point will be invoked in two ways.

- Specifying a Fieldbus point name on the Faceplate Property page
- By setting the TagName parameter at runtime.

3.5 Manipulating GUS Fieldbus Faceplates

Changing SP, PV, OP value

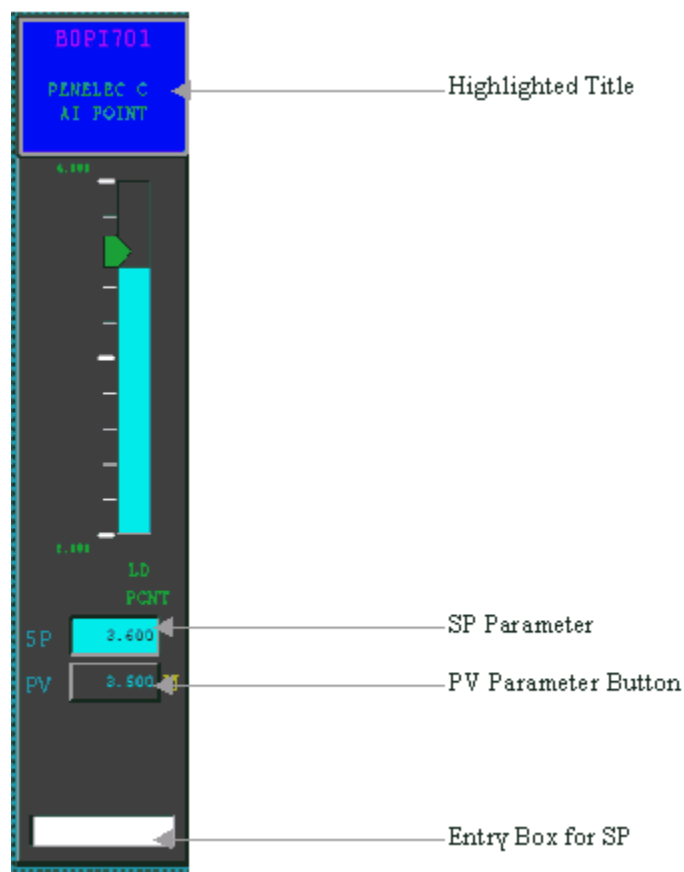
A parameter represented by a button on the Faceplate may be changed under the appropriate conditions. The data owner determines the access permissions for changing any parameter. Not all buttons appear on all Faceplate displays.

Step	Action
Select the Parameter	
1	Press the SP or OP key on the keyboard. Or Touch the screen over the button representing the parameter to be changed. Or Click the left mouse button when the cursor is over the button representing the parameter to be changed. RESULT: - The background of the entry box at the bottom of the Faceplate changes from black to white, and a cursor appears, positioned for data entry. - The parameter button reverses video; that is, the background changes from black to cyan, and the text changes from cyan to black. - The background of the upper Title Area of the Faceplate changes from black to blue, indicating the Faceplate is selected.
Enter a Value	
2	Type a value when the entry box has a cursor. Or Press the Raise/Lower or the Fast Raise/Fast Lower keys on the keyboard. (You can press these keys continuously to ramp or soak the value.) NOTE: To clear an incorrect value, press the corresponding SP, OP, or Clr Entr / numeric-keypad * key in the IKB or OEP, enter the correct value, and press Enter. To clear a single incorrect character, press the backspace key once.
Deselect the Parameter	
3	After typing a value, press the Enter key to deselect the parameter.

Faceplate – parameter value change

The following figure shows a Faceplate selected for a parameter change.

Figure 34 Parameter Value Change



Changing Mode

The Mode button appears below the SP/PV/OP button set, with the current mode appearing as the label.

Step	Action
1	Touch the screen over the Mode button. Or Click the left mouse button when the cursor is over the Mode button. RESULT: • The Mode button reverses video, the background changes from black to yellow, and the text changes from yellow to black. • The selections for mode appear as small buttons in the Graphic Area along the left-side of the analog bar for analog-related points, and above the button set for digital points. Fieldbus will show a column with a maximum of seven buttons, with the four-letter character name for each mode. • The current mode has its text color as cyan. The other mode selections have green as the text color.
2	Click the left mouse button when the cursor position is over the OOS, MAN, AUTO, CAS, NORM, ROUT, OOS, or RCAS button, and then press the Enter key or select the OK button to confirm the choice. Or Touch the screen over the small button representing the mode selection, and then press the Enter key or select the OK button to confirm the choice. Or Press the MAN, AUTO or NORM key on the IKB or OEP keyboard.

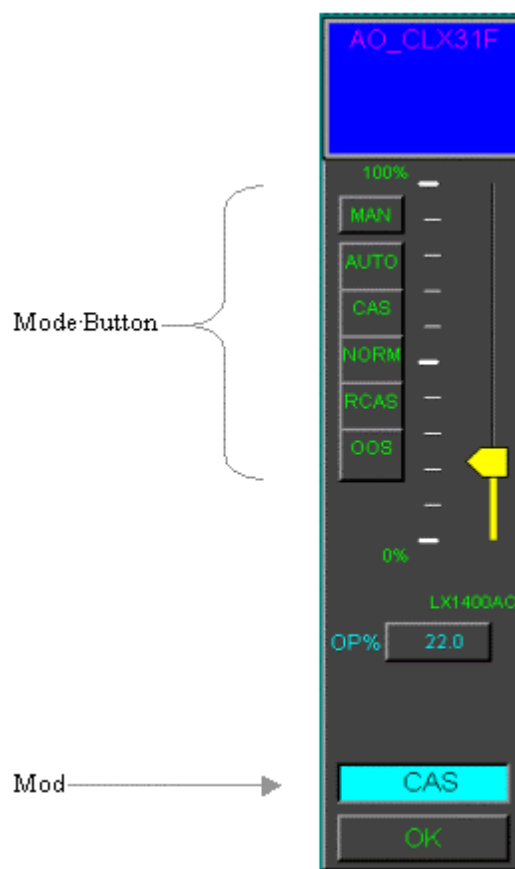
Deselecting mode change

Pressing the Enter key or the OK button to confirm the Mode change automatically deselects the Mode change. If a timeout period expires or you begin another operation, this also automatically deselects the mode change.

Faceplate – mode change

The following figure shows a Faceplate selected for a mode change.

Figure 35 Mode Change



Changing Mode Attribute

The Mode attribute indicates who has the right to change the parameters of a point. The Mode Attribute has two possible values:

- **Operator** – The operator can change the SP, OP, Mode, ratio, and bias of a point. If the Mode attribute is operator, the indicator is a blank.
- **Program** – A program can provide the SP, OP, mode, ratio, and bias of the point. The indication on the display for the Mode attribute is a “P,” for “Program,” displayed preceding the mode name on the Mode button.

Mode Attribute appears as a selection button for non-Fieldbus Analog Input, Analog Output, and PID types.

For points that have a Mode attribute, it is also possible to change the Mode attribute while selecting a Mode.

To change the Mode Attribute, do the following:

Step	Action
1	Click the left mouse button when the cursor is over the PROG button, and then click the left mouse button when the cursor is over the OK button. Or Touch the screen over the PROG button, and then touch the OK button. Or Press NORM on the IKB or OEP keyboard.

Error Handling

Access errors

If you attempt a change when the key level access does not permit the change, a message displays indicating “ACCESS LEVEL ERR”.

If you attempt to change a parameter and the mode does not allow it, a message displays indicating “READ ONLY PAR.”

Input Errors

The default error handling for Faceplate is: Do not change a value if the following is true:

- The value entered is beyond the upper or lower limit for the parameter, or
- The content of the value is not compatible with the value type, such as non-numeric characters when only numeric characters are appropriate.



ATTENTION

There is no user-visible indication of an input error unless you script the OnError event. The OnError event occurs for any error raised by the data owner.

Invalid Tagname error

The error reporting Invalid Tagname will be reported on the Faceplate itself. The OnError event will not occur for Invalid Tagname.

3.6 Fieldbus Modes

General description

Fieldbus Mode management and the set of possible values for the Mode differ from the traditional TPS Mode.

The following applies to Fieldbus points:

- The Mode is presented as a composite of the *actual* and *target* Modes of the CM point.
- The number of parameter values for the CM Mode is extended to eight.
- Not all modes are valid for all point types.

Mode definitions

The following table describes the Fieldbus modes.

Table -19 Fieldbus Modes

Mode		Applicable Points	Meaning	Service Priority
2 Chars	4 Chars			
OS	OOS	AI, AO, PID	Out of Service	8
IM	IMAN	AO, PID	Initialization Manual	7
LO	LO	AO, PID	Local Override	6
M	Man	AI, AO, PID	Manual	5
A	Auto	AI, AO, PID	Automatic	4
C	Cas	AO, PID	Cascade	3
RC	Rcas	AO, PID	Remote Cascade	2
RO	ROut	PID	Remote Output	1

Mode attributes

The Fieldbus mode attributes are:

Actual – the current value of the mode.

Target – the value that the operator wants for the mode.

Permitted – the allowable target modes.

Normal – the mode (or combination of modes) to which the target will be set when the operator presses the NORMAL key.

Permitted modes apply to the target mode. A write request to the target is rejected if it does not match the permitted list.

The Actual mode is not constrained by the Permitted mode, because some modes are required for initialization

Displayed Mode values

When the *actual* and *target* modes are the same, the Faceplate will represent them as a single mode.

When the actual and target modes are different, Faceplate will display them as a combination mode, as indicated in the following table.

Table 3-20 Combination Modes

Actual	Target
M	(C)
LO	(A)
MAN	(A)
AUTO	(C)
CAS	(RC)



ATTENTION

IMan and LO are not legal target modes.

The following table lists the normal modes that can be displayed for a Fieldbus point:

Table 3-21 Normal Modes

Mode		Meaning
2-Chars	4 Chars	
M	Man	Manual
A	Auto	Automatic
C	Cas	Cascade
RC	Rcas	Remote Cascade
RO	Rout	Remote Output



ATTENTION

OOS and LO are not valid normal operating modes.

3 GUS Faceplate Control

3.6 Fieldbus Modes

4. Standard Display Applications

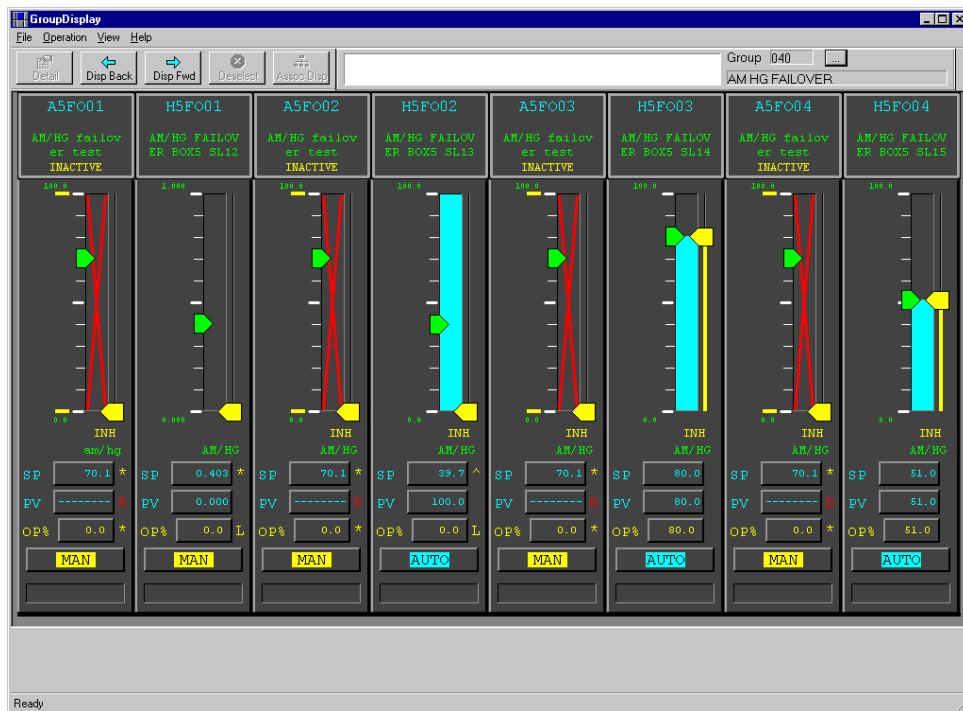
4.1 GUS Group Display

General description

The GUS Group Display looks similar and behaves similar to the Native Window standard Group Display. An example is shown below.

If a point has tuning parameters, those parameters will appear in the area below the group of eight Faceplates when you select one of the Faceplates. Currently tuning parameters do not display for Fieldbus points.

Figure 36 GUS Group Display



4 Standard Display Applications

4.1 GUS Group Display

Runtime behavior

- Only one GUS Group Display can be running per GUS node. Only three faceplates can run if the GUS Group Display is running.
- When Faceplate in the Group Display has focus, it interacts with the IKB/OEP keys. You may interact with the Faceplate using the point manipulation keys: SP, OUT, MAN, AUTO, NORM, and rampkeys.
- Errors are shown in the message area on the toolbar at the top of the display.
- The display size is 1024 x 768. The display does not zoom-to-fit.
- The display registers itself with SafeView with the category "GroupDisplay."
- The display window has the full functionality of a window (for example, it can be closed and it can be moved).
- Because it is a well-behaved client of SafeView, the GUS Group Display can be constrained by the properties set by the SafeView workspace.

GUS Group display buttons

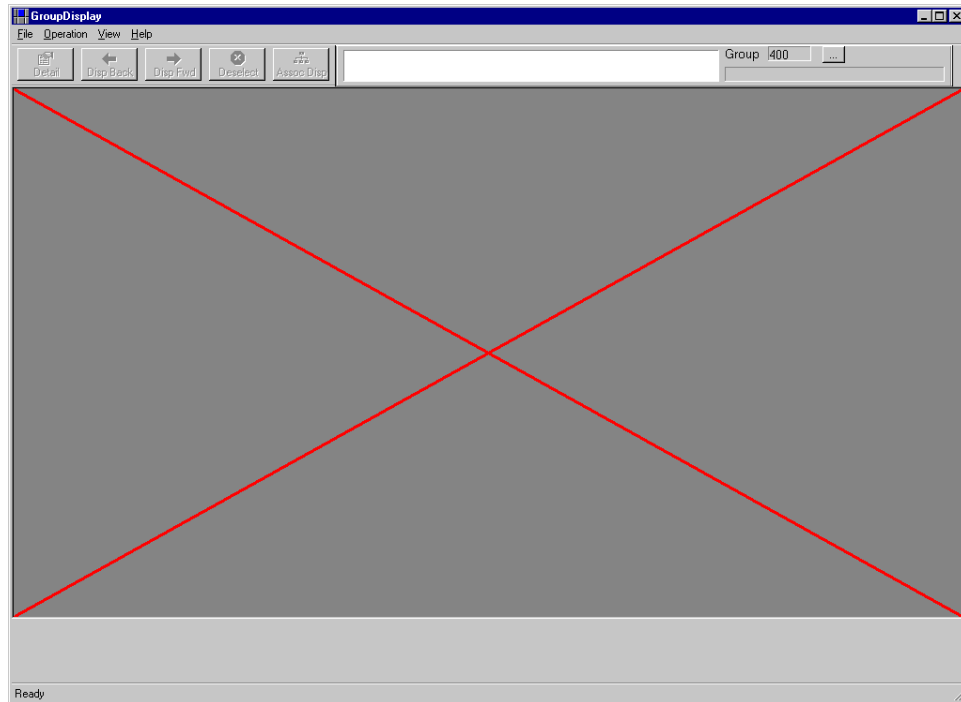
The following table describes buttons located at the top of the GUS Group Display. These buttons do not appear on the Native Window Standard Group Display.

Table 4-1 GUS Group Display Buttons	
Button	Description
Detail	Invokes the Detail display for the selected point.
Disp Back	Invokes the Group previous in numerical order to the displayed group.
Disp Fwd	Invokes the Group subsequent in numerical order to the displayed group.
Deselect	Deselects the selected faceplate slot.
Assoc Disp	Invokes the Associated Display for the selected point or the Group if a point is not selected.
Help Disp	Invokes the Help Display for the Group.
...	Invokes a dialog box to enter a group number.

TPN Connection error

When the Group Display fails to connect to the TPN or loses its connection to the TPN, a red “X” is drawn across the Group, as shown in the following figure.

Figure 37 GUS Group Display With No Connection to TPN

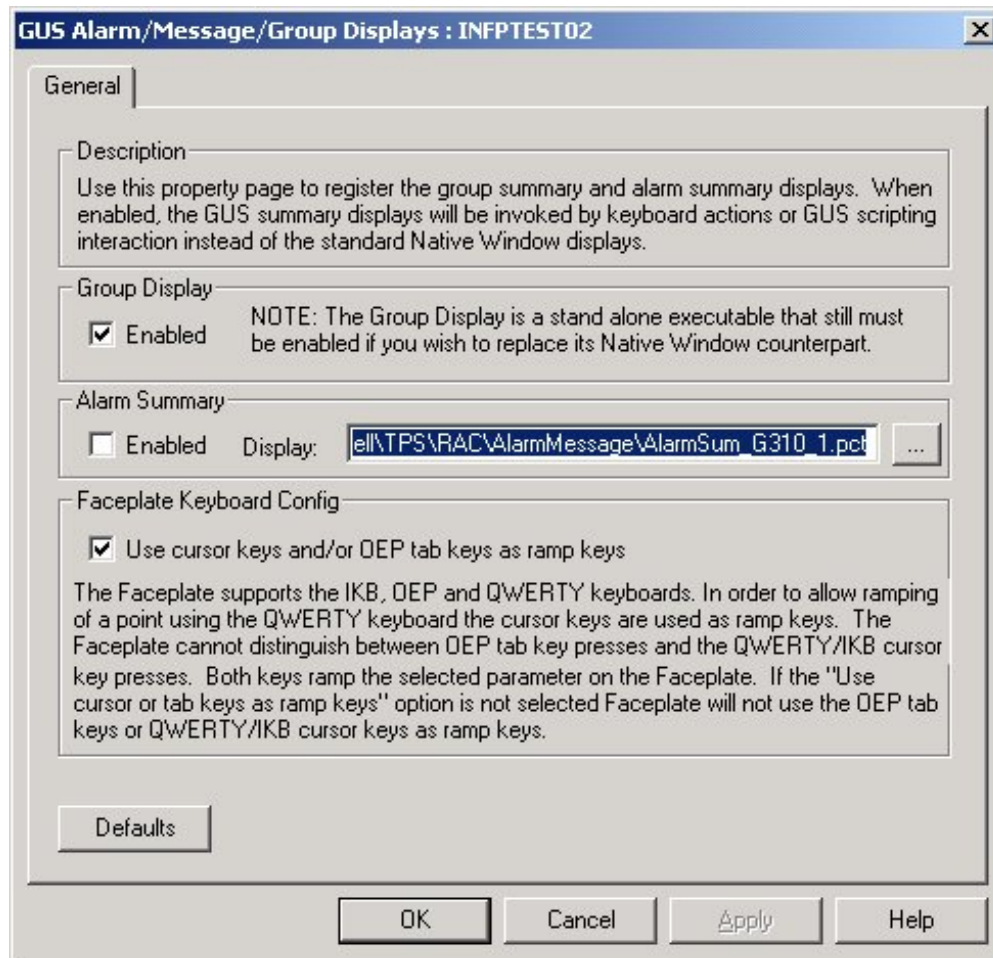


4.2 Configure GUS Group Display to be Standard Group Display

Perform the following steps to configure the GUS Group Display to be the Standard Group Display for a GUS node.

Step	Action
1	Invoke the Configuration Utility .
2	Select GUS Alarm/Message/Group Displays from the Configure menu. RESULT: The GUS Alarm/Message/Group Displays configuration page is displayed.
3	Check Enabled for the Group Display.

Figure 38 GUS Alarm/Message/Group Displays Configuration Page



4 Standard Display Applications

4.3 Invoking the GUS Group Display

4.3 Invoking the GUS Group Display

If you have configured the GUS Group Display to be the standard Group Display, then you may invoke the display from the GROUP Key or from a GUS script.

Invoking a Group from a Native Window display will display the Native Window Group Display.

To invoke the Group Display from a GUS script, use the GROUP Actor:

```
Sub OnLButtonClick()  
    Group 40  
End Sub
```



ATTENTION

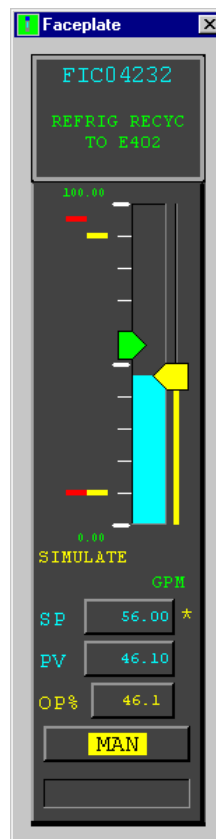
A slot to be selected upon invocation cannot be passed through the GROUP Actor to the GUS Group Display.

4.4 Faceplate Application

The GUS faceplate OCX has been embedded in a standalone application, faceplate.exe. All running GUS displays manipulate the same faceplate. By being a standalone application, the Faceplate Application allows the maximum number of GUS displays to be available to the user. It also conserves display space by allowing the faceplate to exist outside of a GUS display. The size of the faceplate window is 160 x 600.

This Faceplate Application (faceplate.exe) can be created and manipulated using GUS scripting. You cannot run the faceplate application from a command line or by double clicking on the file in Windows Explorer.

Figure 39 GUS Faceplate



4 Standard Display Applications

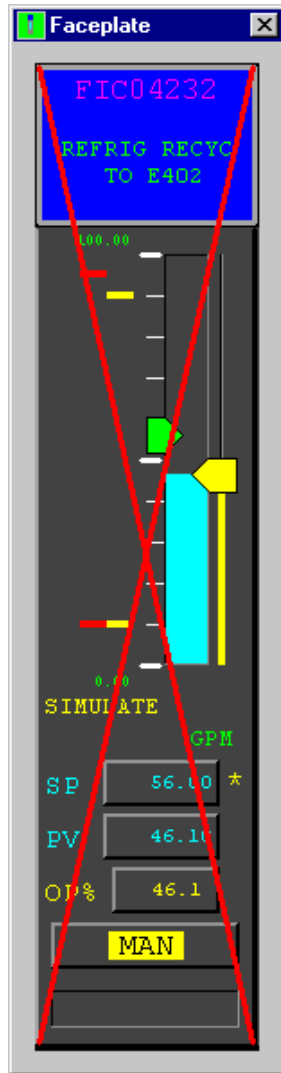
4.4 Faceplate Application

Runtime Behavior

The following is a description of the behavior of the Faceplate Application during runtime.

- Only one faceplate.exe may be running per GUS node. All running GUS displays manipulate the same faceplate application.
- Setting the tagname to an empty string will blank the faceplate, but the faceplate window will still be visible.
- When the faceplate window has focus, it interacts with the IKB/OEP keys. You can interact with the faceplate using the point manipulation keys, SP, OUT, MAN, AUTO, NORM, and rampkeys.
- The Faceplate Application's default error handling is to display a message box containing the error message. The message box must be acknowledged before you can continue working in the faceplate.
- The Faceplate Application registers itself with SafeView with the category "Faceplate."
- The faceplate window has the full functionality of a window (for example, it can be closed and it can be moved). Because it is a well-behaved client of SafeView, it can be constrained by the properties set by the SafeView workspace.
- The Faceplate Application will shutdown when there are no displays (clients) connected to the faceplate, and you have closed the faceplate window. You can close the faceplate window at any time, but if the faceplate application still has a display (for example, a client) connected to it, the application will continue to run in the background.
- If the Faceplate Application fails to connect to the TPN or loses its connection to the TPN, a red "X" is drawn on the faceplate.

Figure 40 GUS Faceplate When There is No TPN Connection



4 Standard Display Applications

4.5 Scripting a GUS Display to Interact with the Faceplate Application

4.5 Scripting a GUS Display to Interact with the Faceplate Application

Identifying the Faceplate Application to a GUS Display

Each GUS display that wants to interact with the faceplate application must assign the faceplate application to an object.

```
dim MyFaceplateObject as object  
set MyFaceplate = createobject("Faceplate.Application")
```

For example, create a public object "MyFaceplateObject" In an OnDisplayStartUp script, set MyFaceplateObject equal to the faceplate object.



ATTENTION

This create can be done in each embedded display.

Main Display/Embedded Display Script

```
-----  
  
public MyFaceplateObject as object  
  
OnDisplayStartup()  
  
    set MyFaceplate = createobject("Faceplate.Application")  
  
End Sub
```

Setting the TagName of the Faceplate from a GUS Display

First, you need to know the object to which the faceplate application has been assigned; then you can set the property, **TagName**, of the faceplate to a valid TPN point.

Faceplate Property	Definition	Type	Default Value	Valid Values
TagName	Property that specifies the point-of-interest displayed in the faceplate	String	(Null String)	Valid Point Names

Button1 Script on Main Display/Embedded Display:

```
public MyFaceplateObject as object

Sub OnLButtonClick()
    MyFaceplateObject.TagName = "A100"
End Sub
```

4 Standard Display Applications

4.5 Scripting a GUS Display to Interact with the Faceplate Application

Setting the TagName of the Faceplate and Defining Error Handling

There is a method on the faceplate application that allows you to set the tagname and set an object in the calling display to handle the faceplate error. The method is `SetTagNameHandleErrors`. It takes two arguments. The first argument represents the tagname of type string. The second argument represents the object that will handle the error.

Faceplate Application Method	Definition	Parameter	Return
SetTagNameHandleErrors	Method used to set the tagname in the faceplate and set an object in the calling display to handle the faceplate errors	TagName as string Errorhandler as object	None

Button2 Script on Main Display/Embedded Display:

```
public MyFaceplateObject as object
```

```
Sub OnLButtonClick()
```

```
    MyFaceplateObject.SetTagNameHandleErrors "A100",  
    text2.methods 'text2 is in the same embedded display
```

```
End Sub
```

Text2 Script on Main Display/Embedded Display: Note: The object handling errors must handle the event, `Faceplate_OnError` (message as string).

```
Sub Faceplate_OnError (message as string)
```

```
    me.text = message
```

```
End Sub
```


Honeywell

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